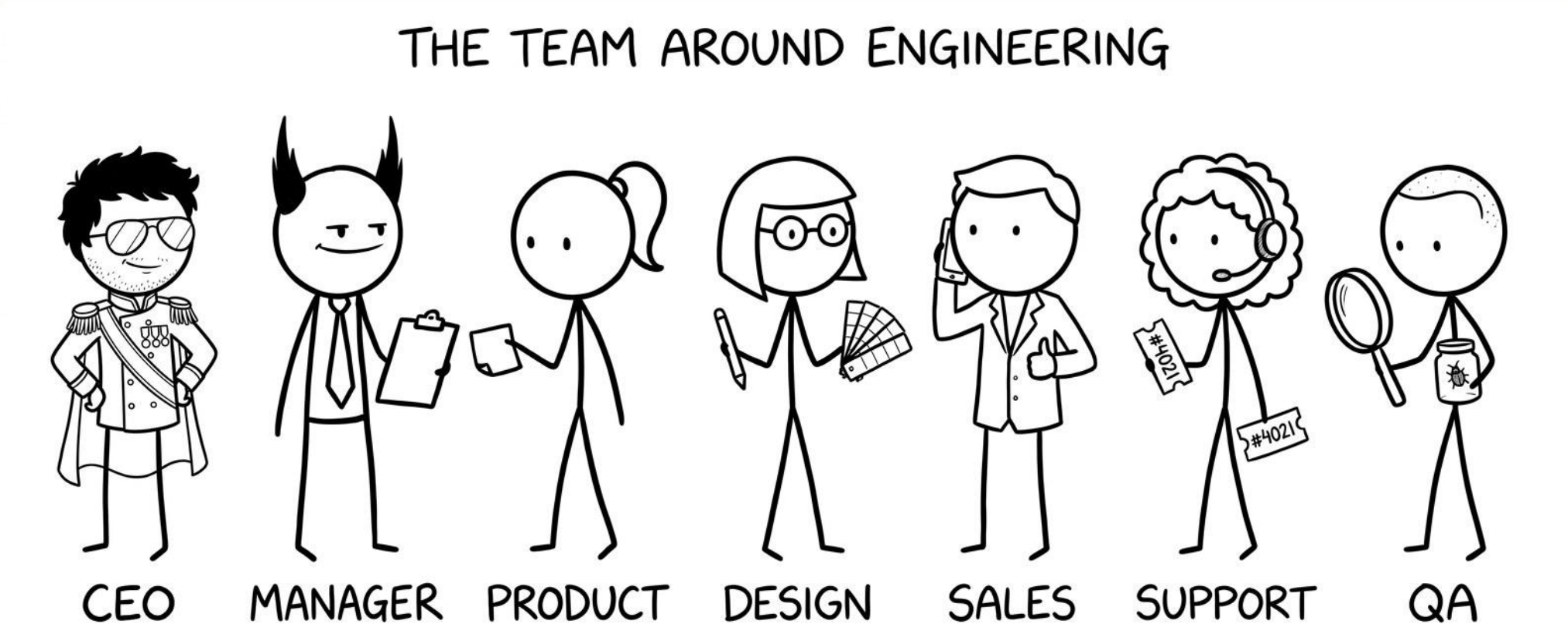
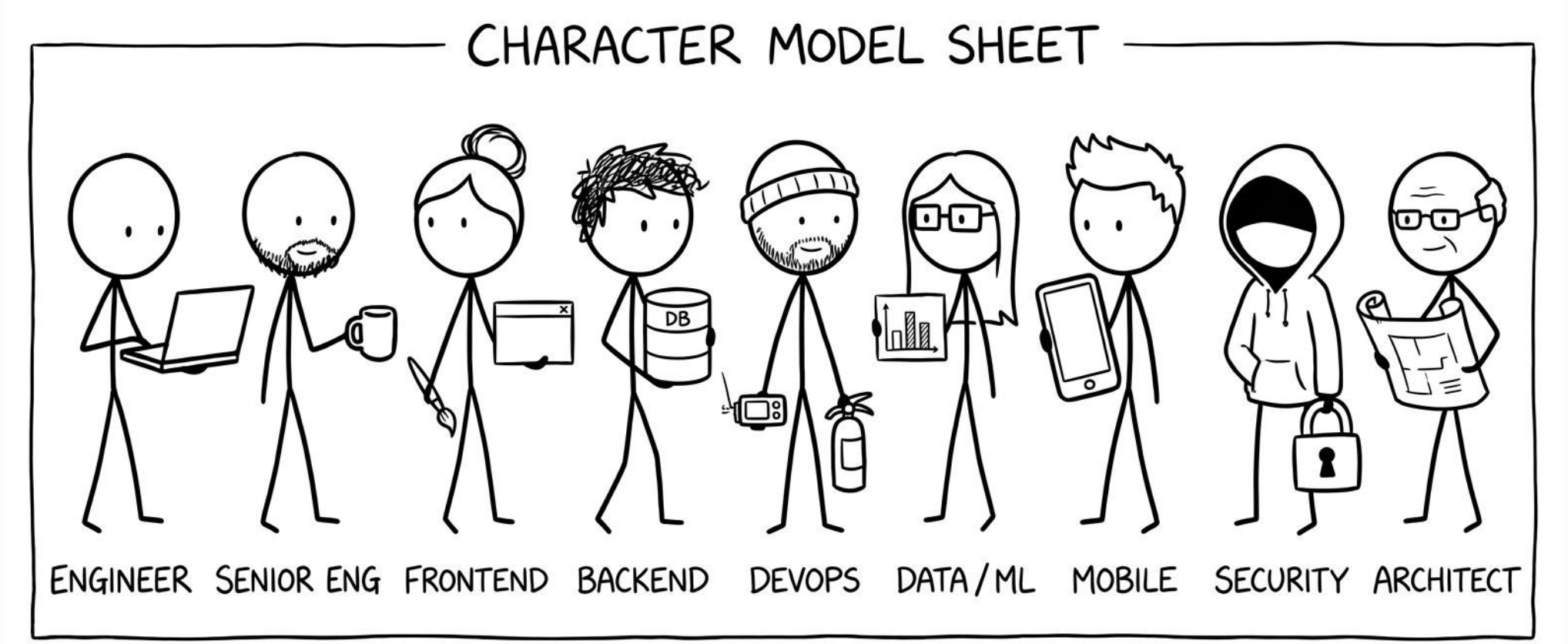
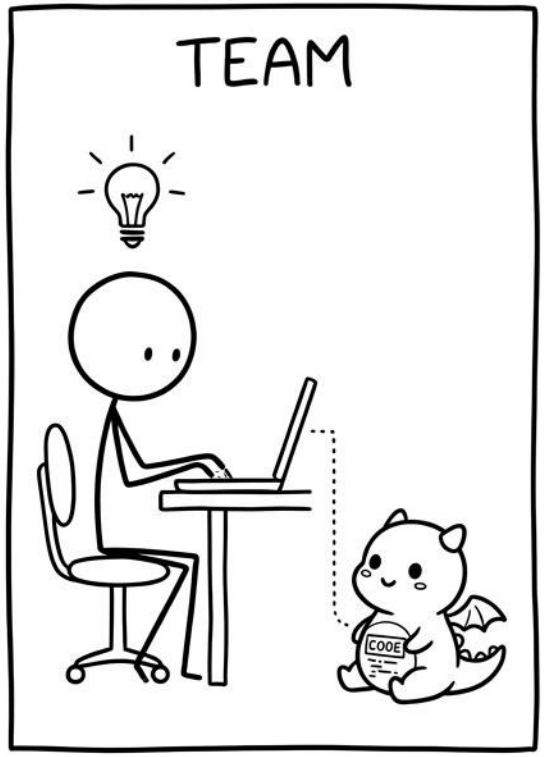
HOW ENGINEERING TEAMS WORK, AND WHY

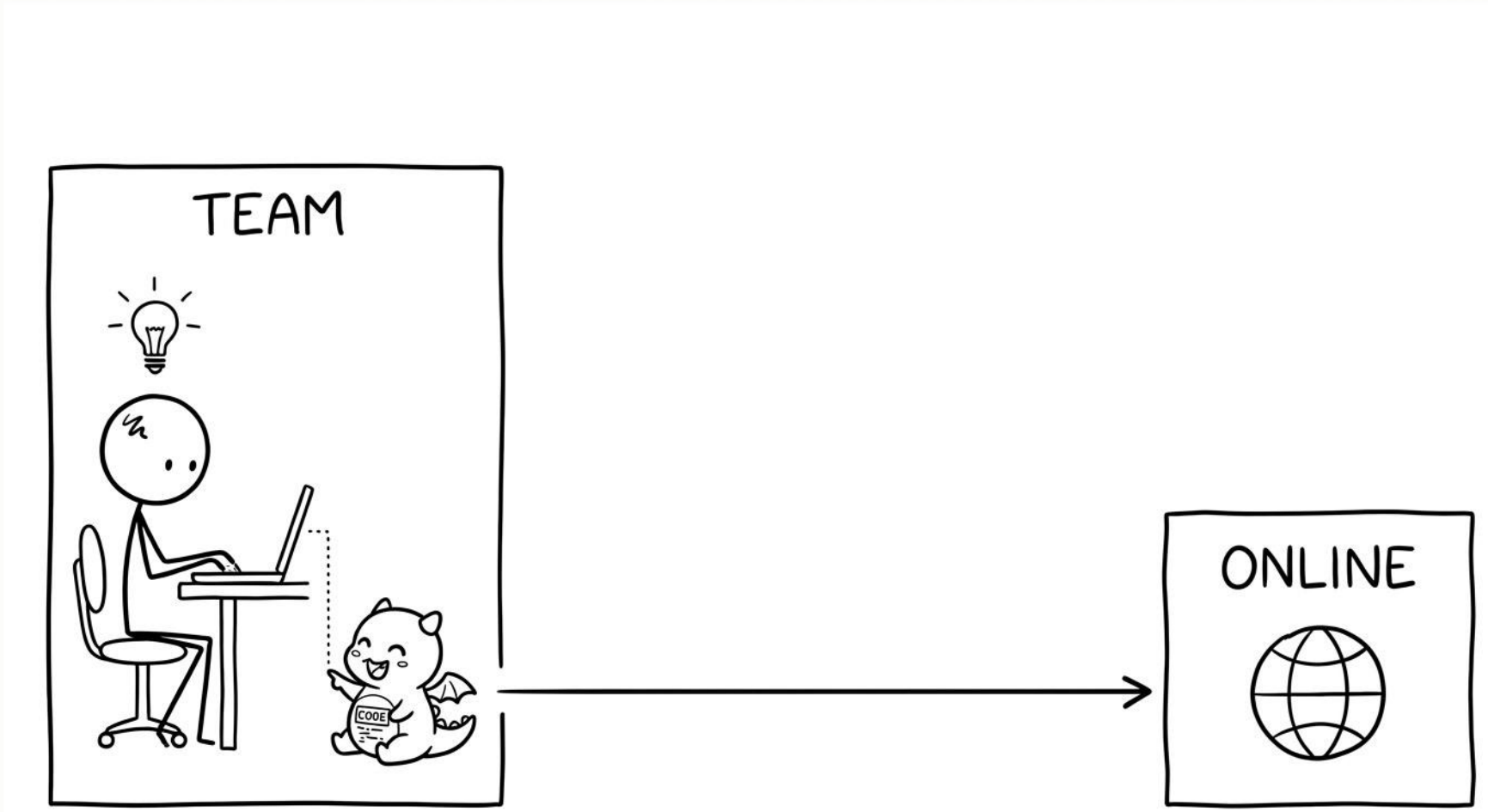
The Evolution of Engineering Teams

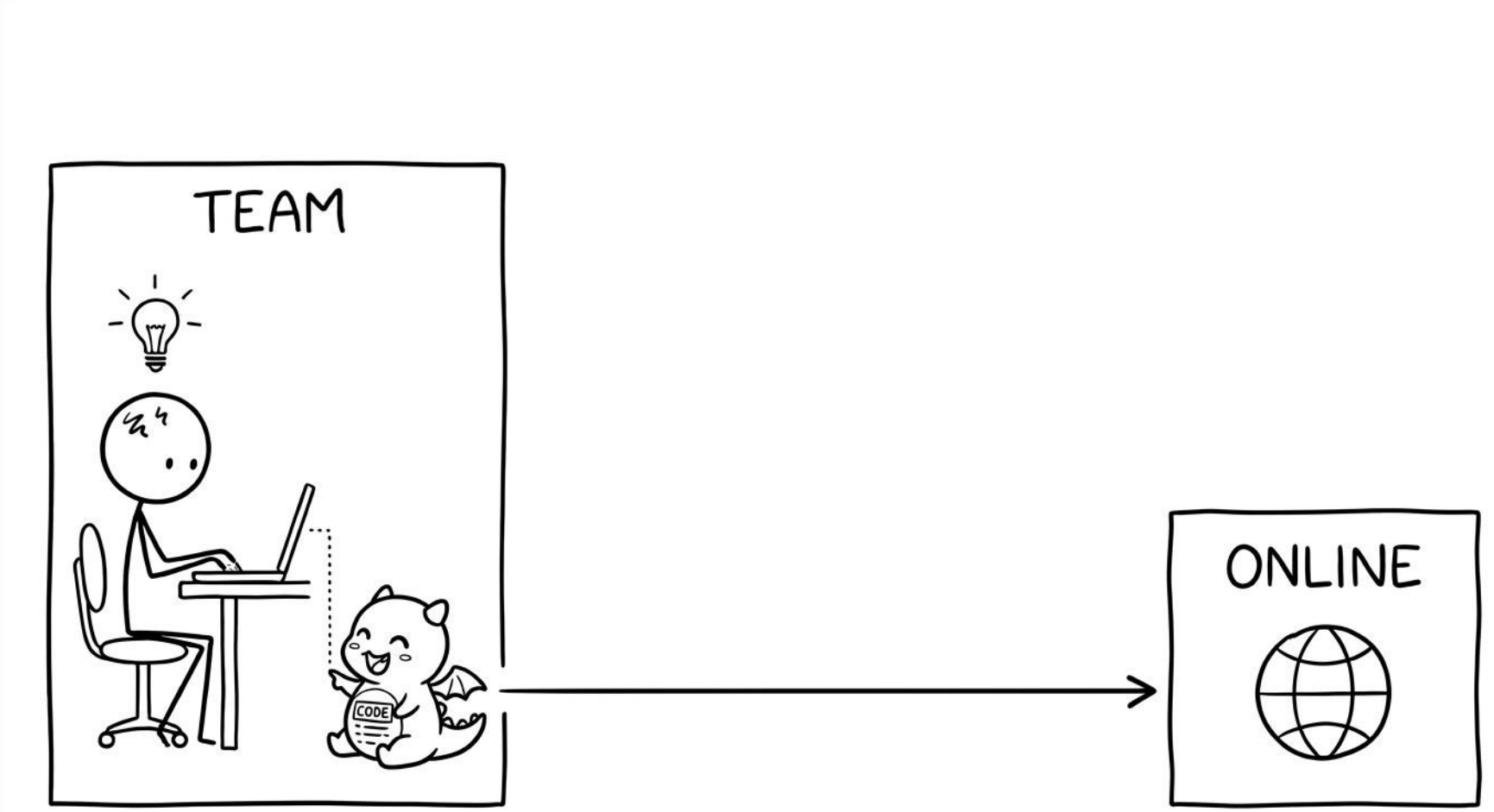
A field guide for non-engineers and new managers, in stick figures.

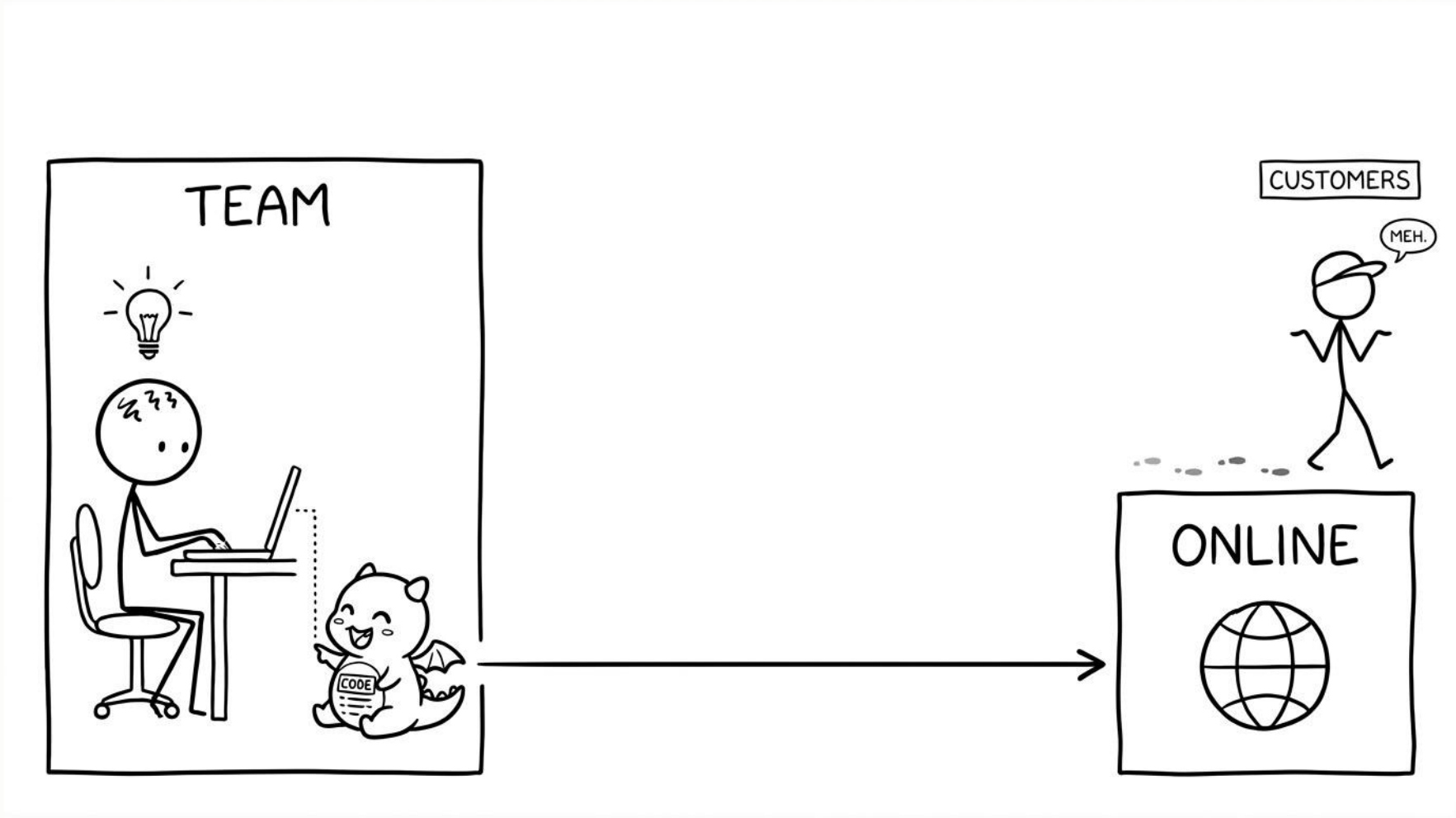

THE TEAM AROUND ENGINEERING
#4021
#4021
CEO
MANAGER
PRODUCT
DESIGN
SALES
SUPPORT
QA

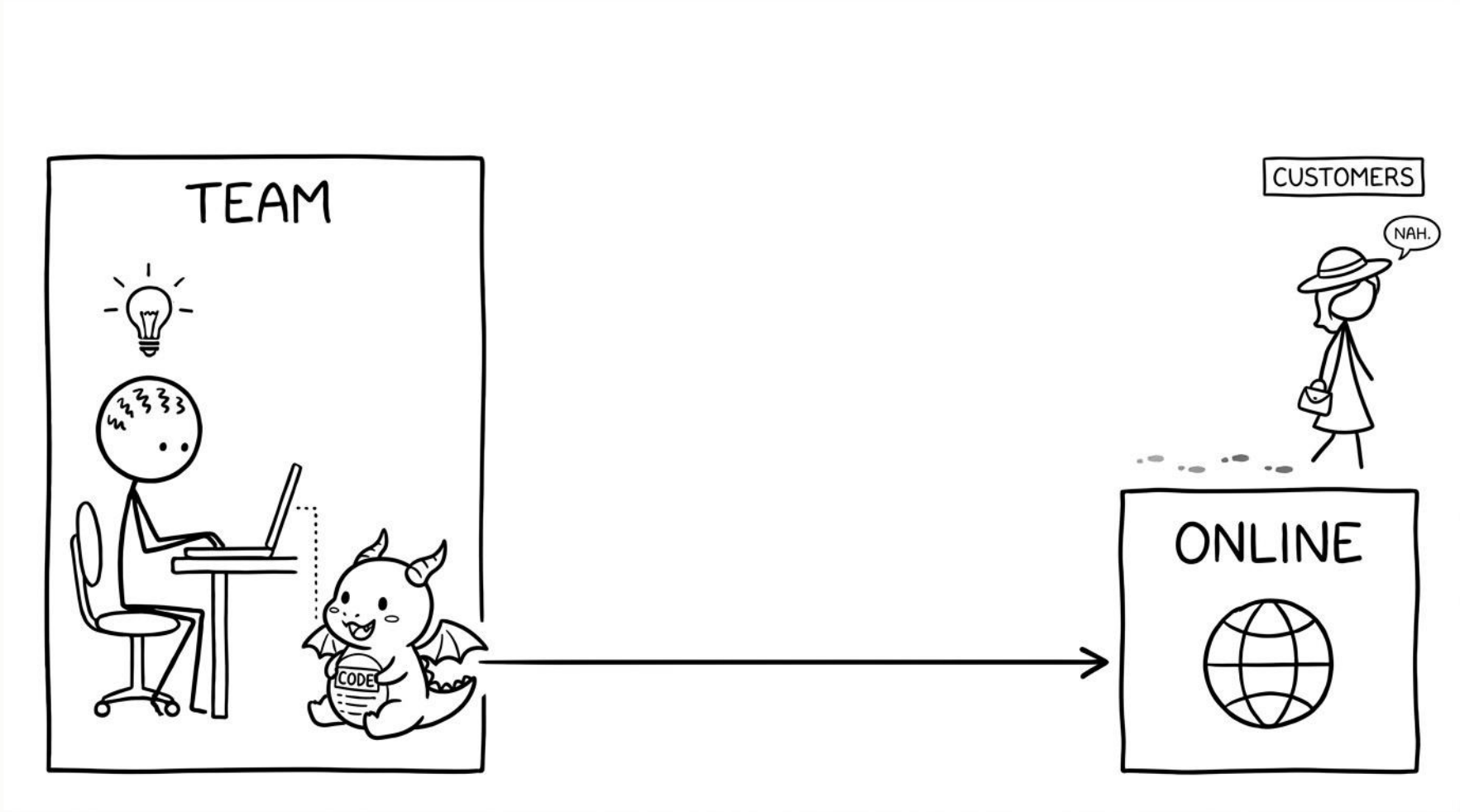

CHARACTER MODEL SHEET
DB
ENGINEER
SENIOR ENG
FRONTEND
BACKEND
DEVOPS
DATA / ML
MOBILE
SECURITY
ARCHITECT

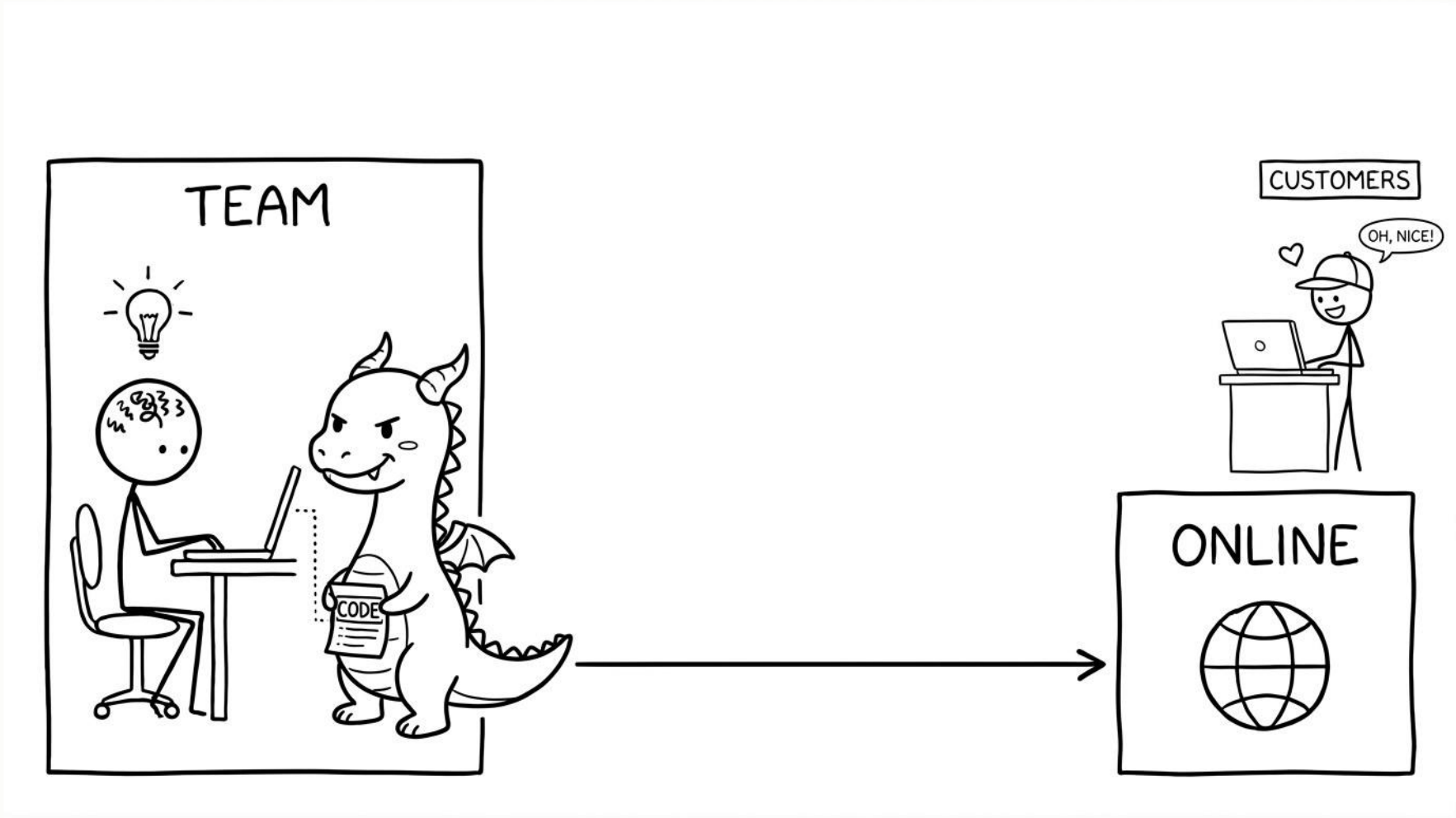

TEAM
COOE

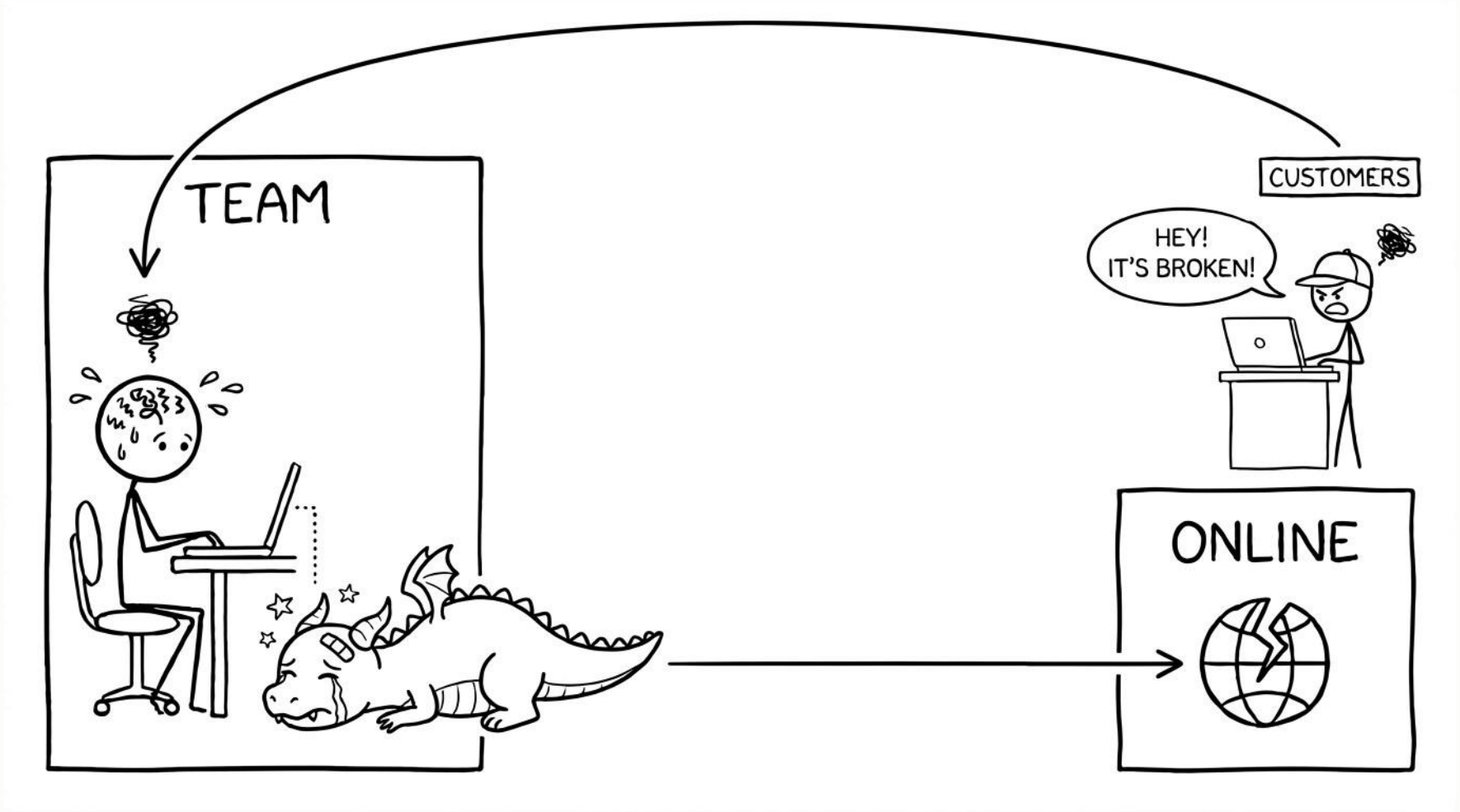

TEAM
CODE
ONLINE

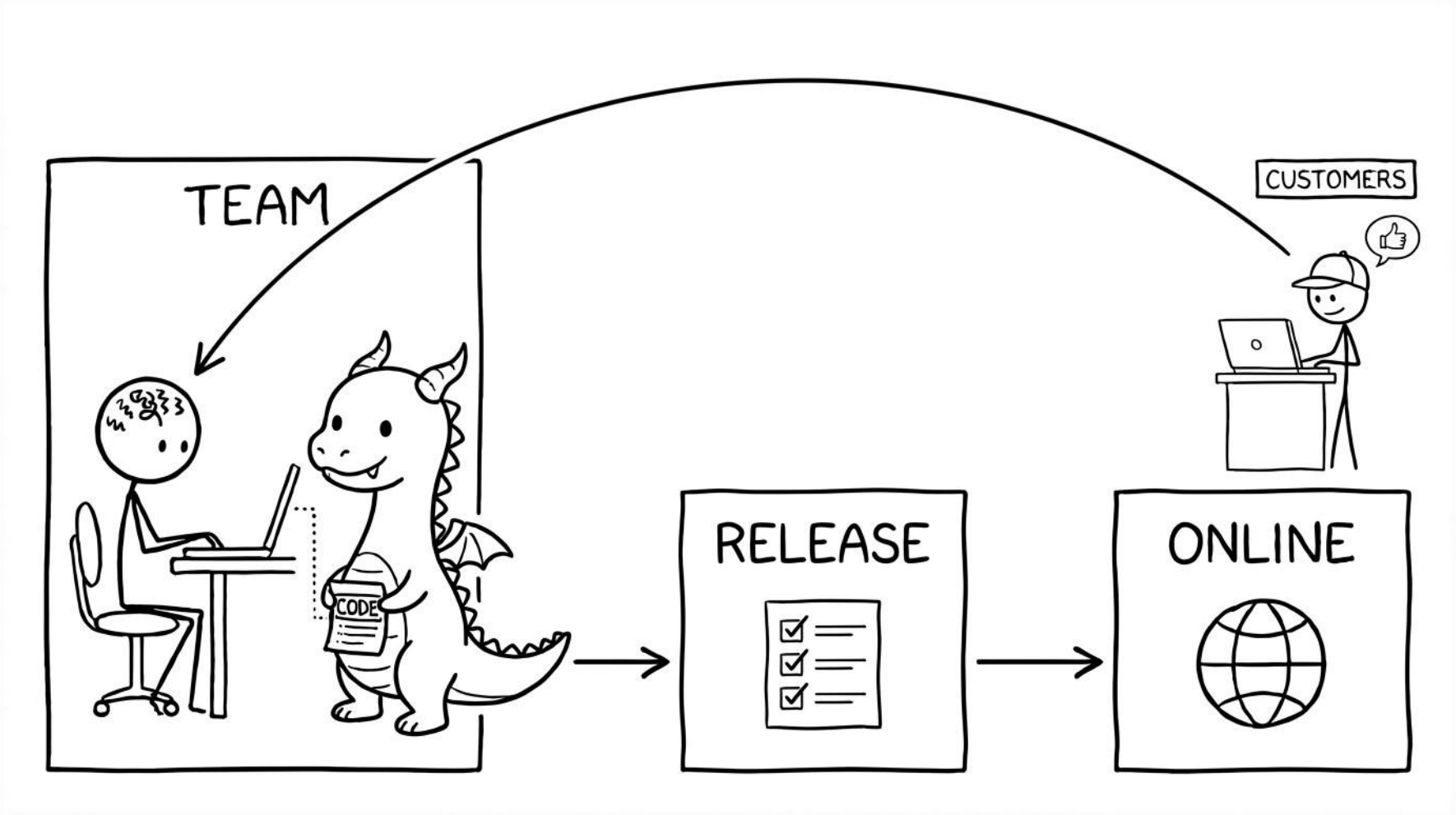

TEAM
CODE
ONLINE

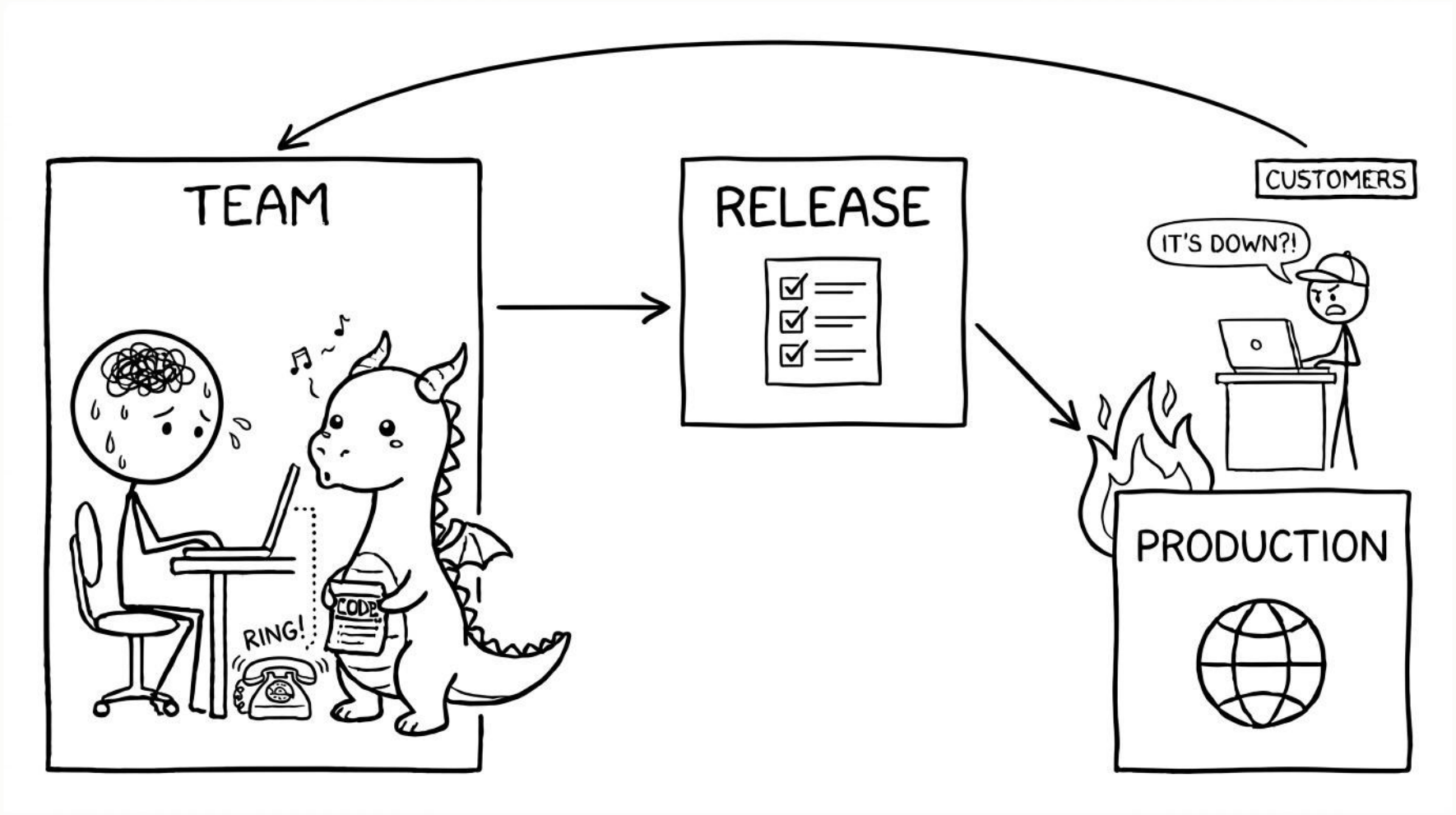

TEAM
CODE
CUSTOMERS
MEH.
ONLINE

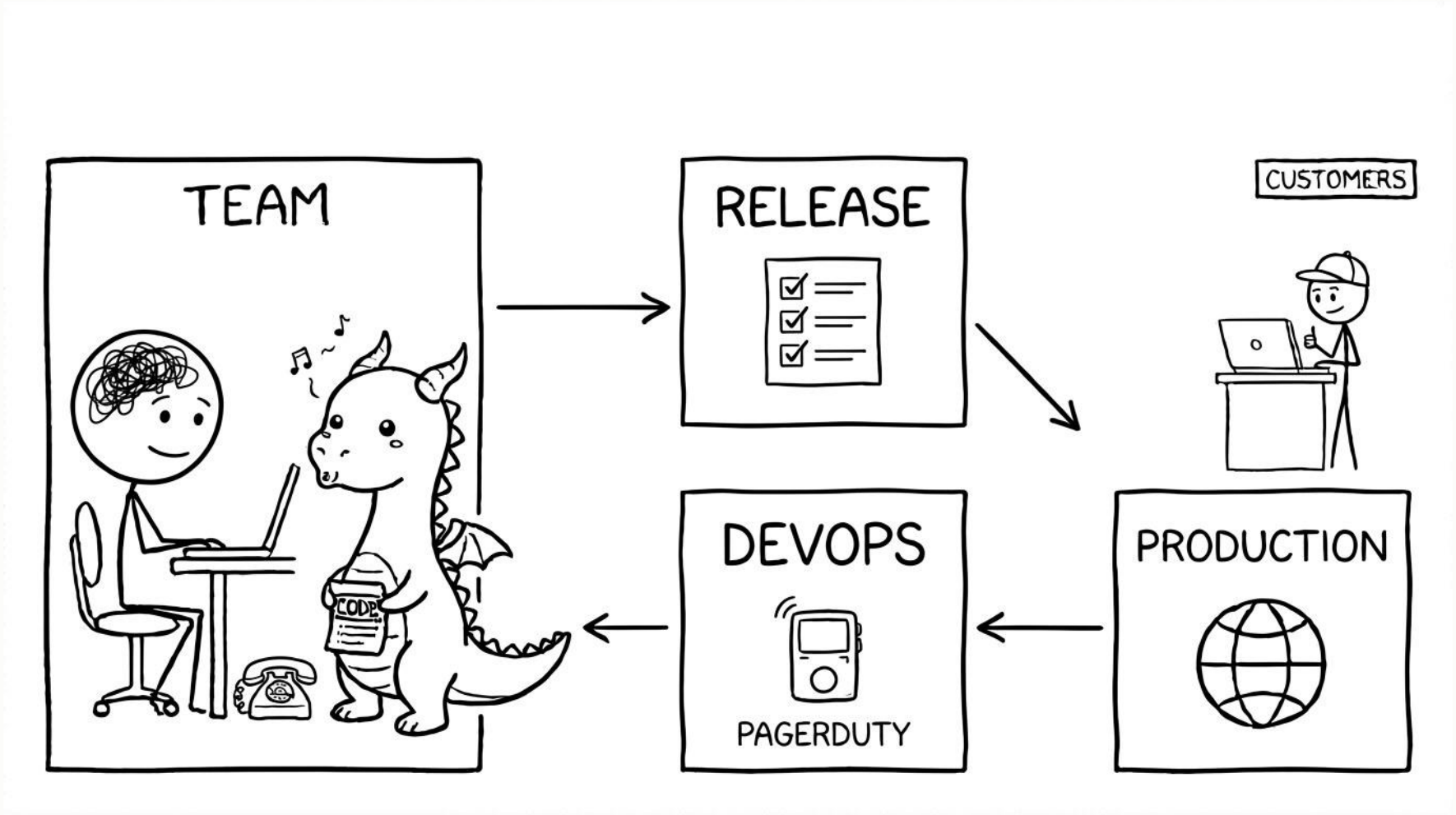

TEAM
CODE
CUSTOMERS
NAH.
ONLINE

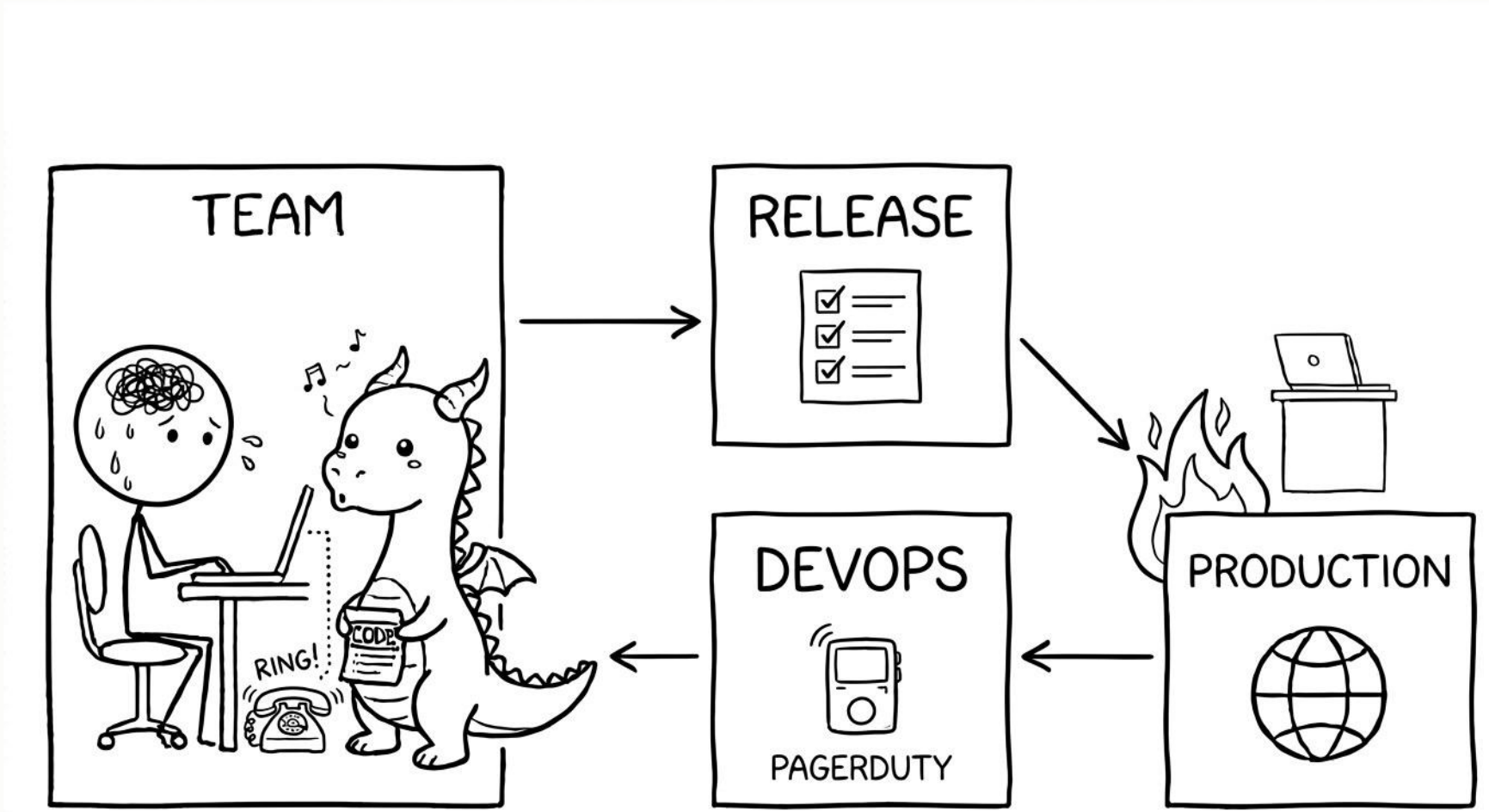

TEAM
CODE
CUSTOMERS
OH, NICE!
ONLINE

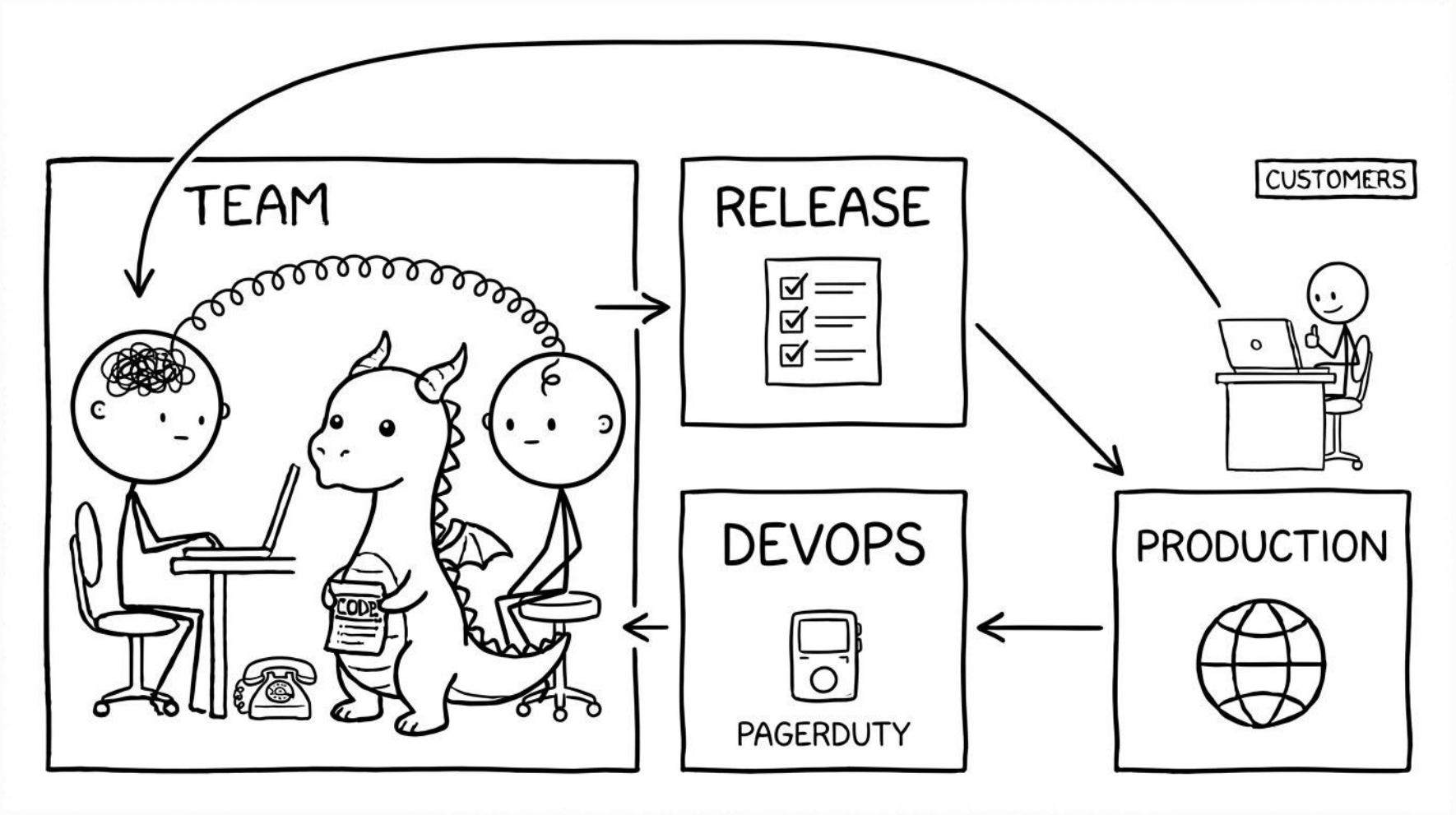

TEAM
CUSTOMERS
HEY!
IT'S BROKEN!
ONLINE

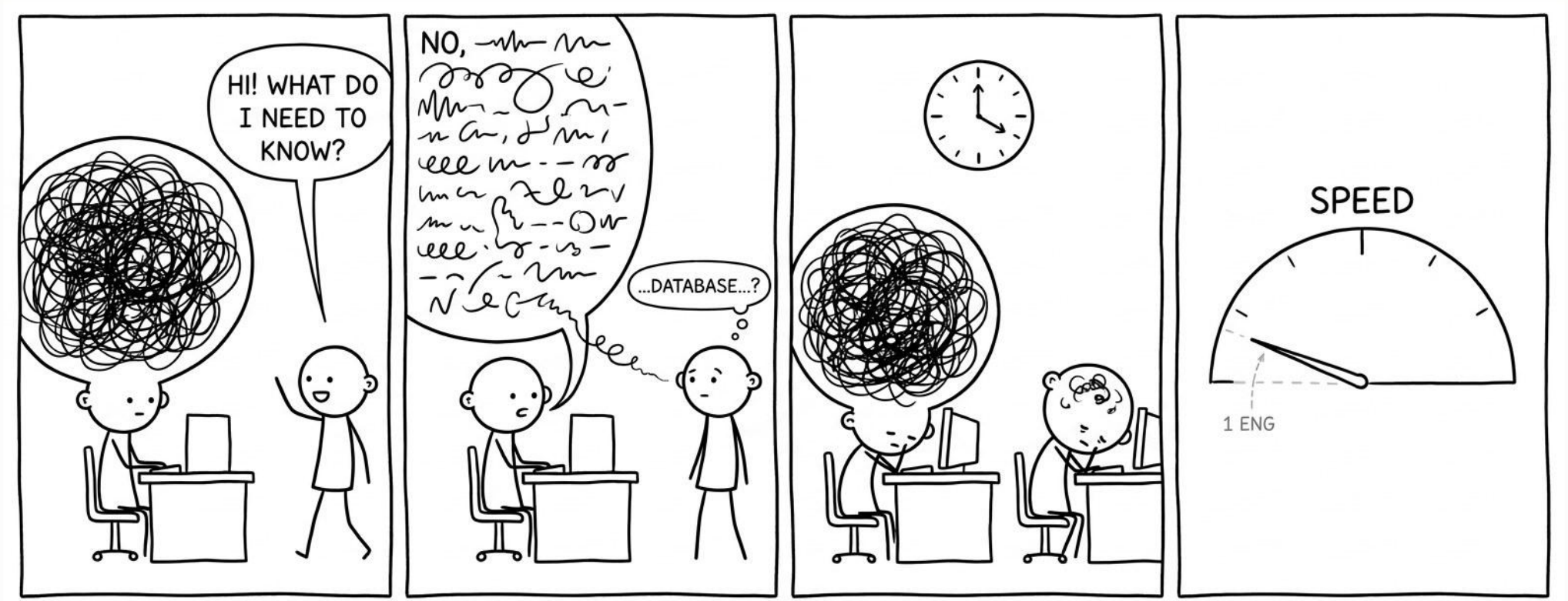

TEAM
CODE
CUSTOMERS
RELEASE
ONLINE


TEAM
RELEASE
CUSTOMERS
IT'S DOWN?!
PRODUCTION
RING!
CODE

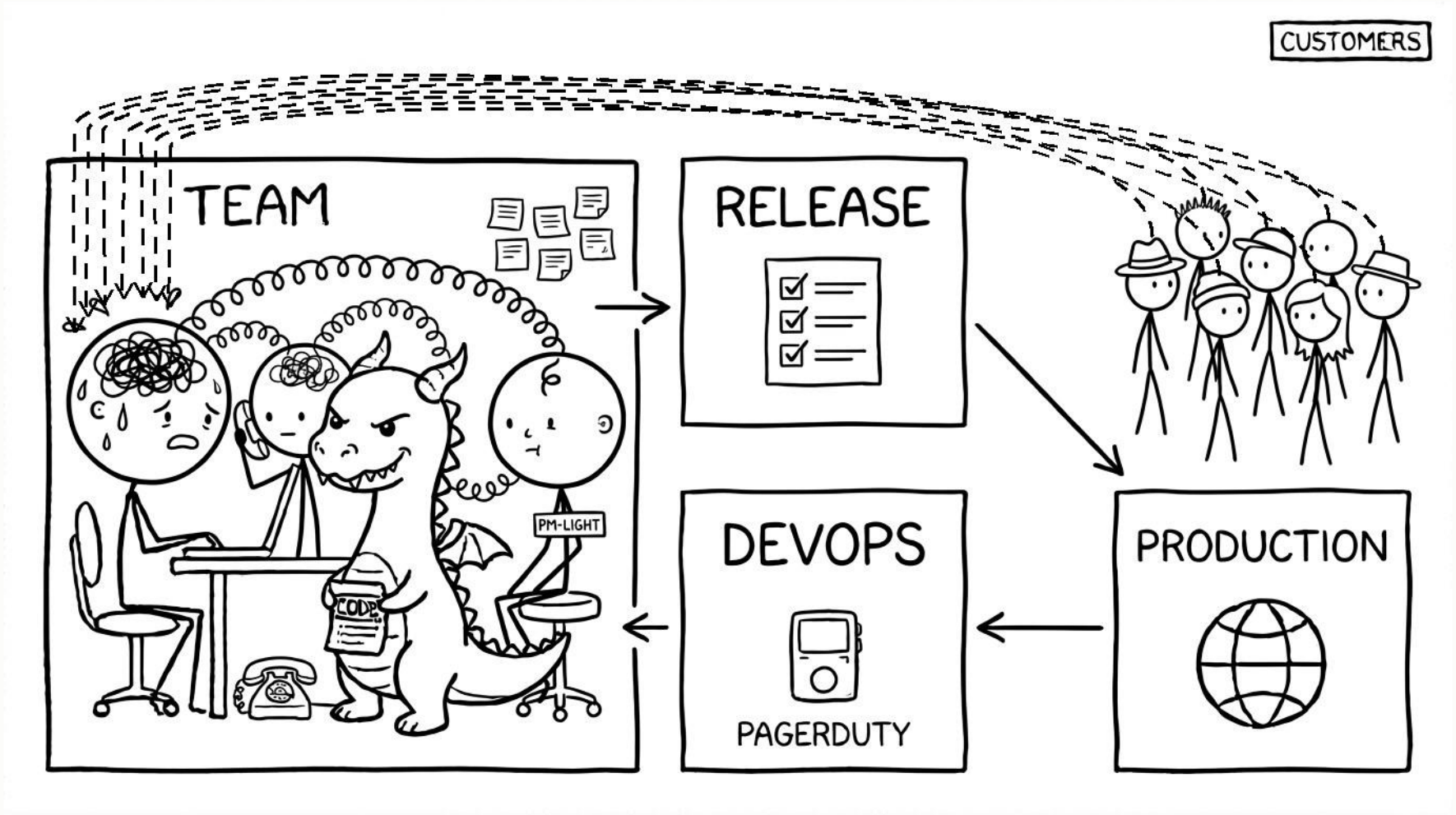

TEAM
CODE
RELEASE
CUSTOMERS
PRODUCTION
DEVOPS
PAGERDUTY

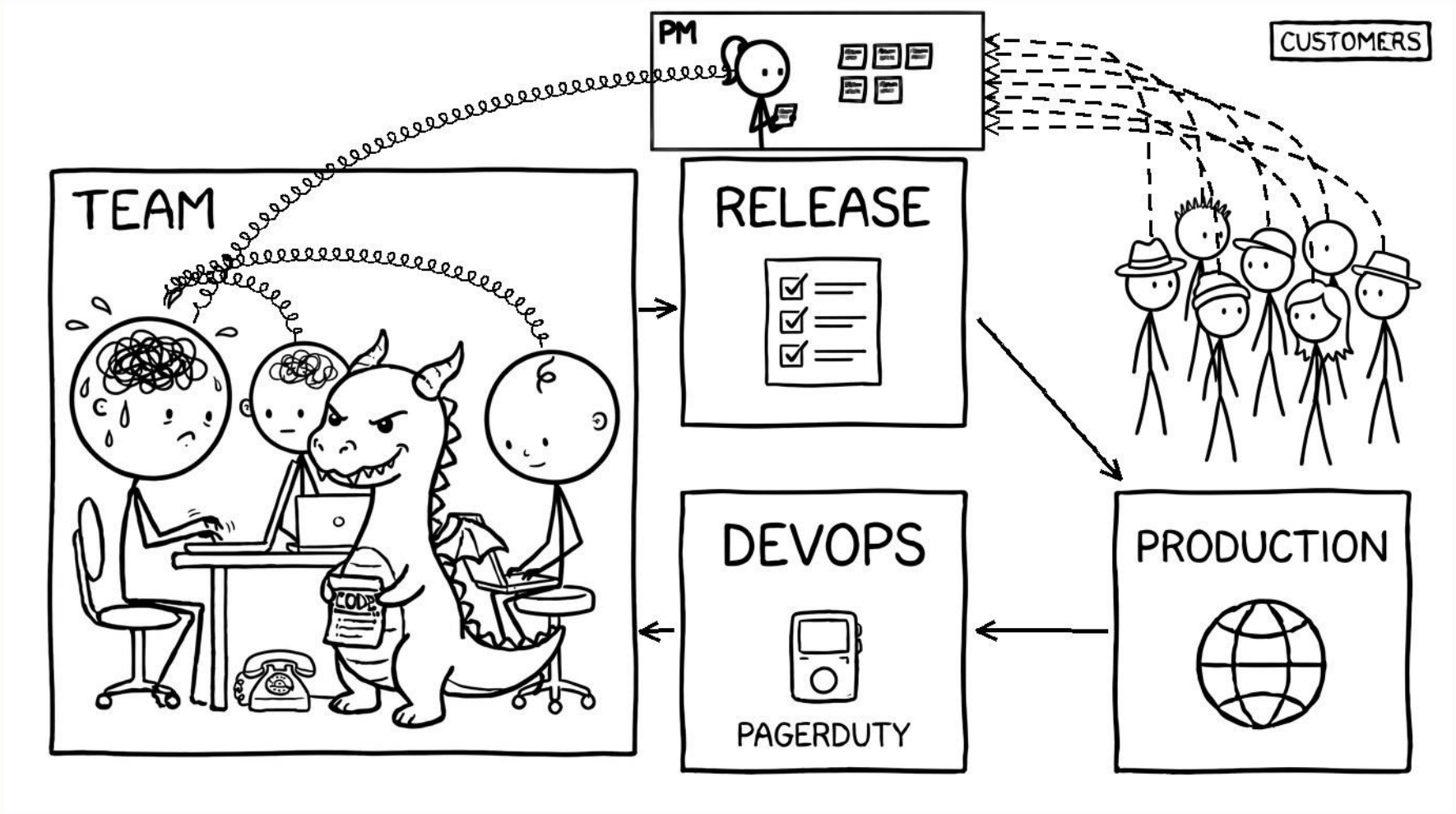

TEAM
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RING!
CODE

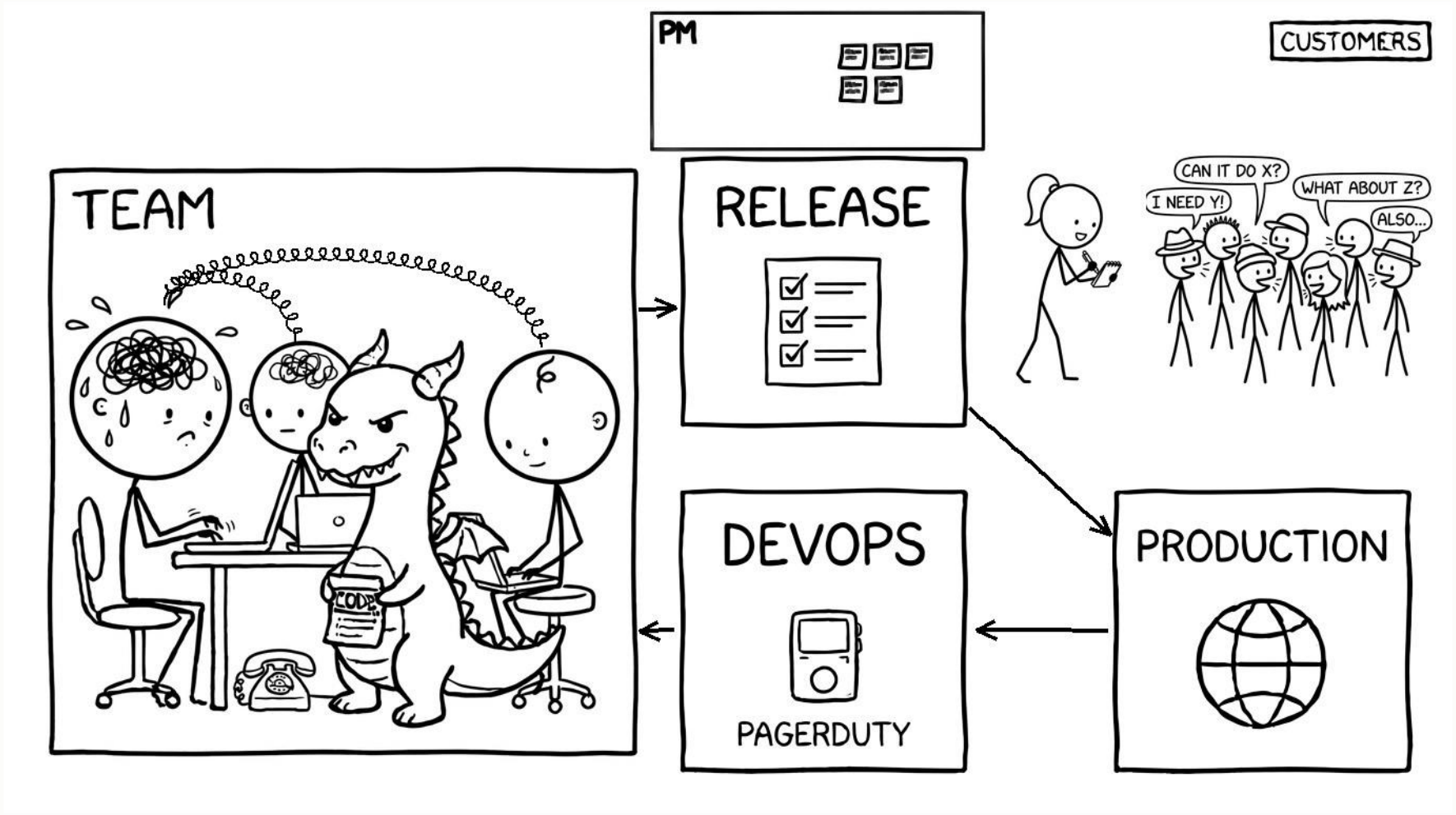

TEAM
CODE
RELEASE
CUSTOMERS
DEVOPS
PAGERDUTY
PRODUCTION

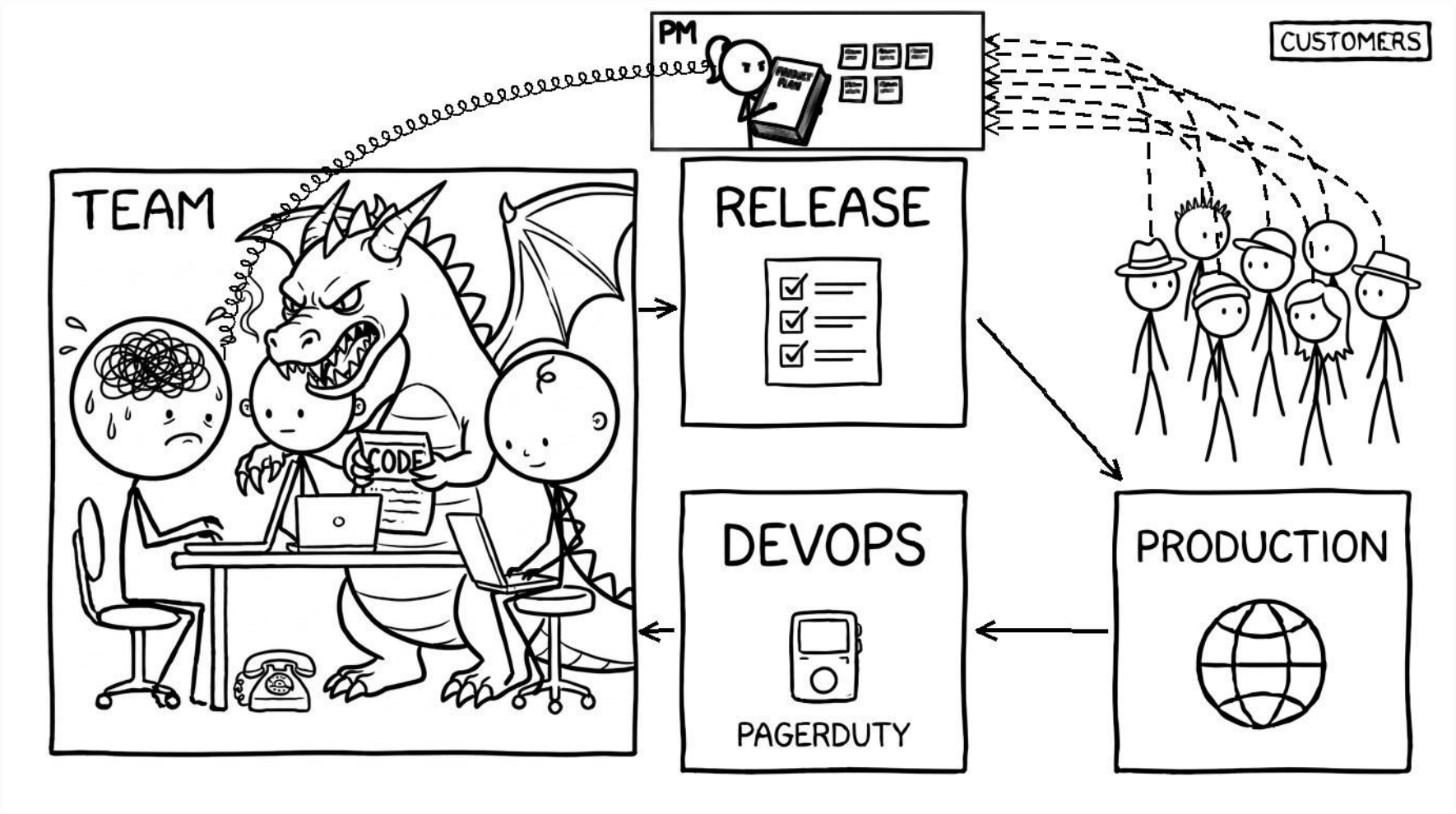


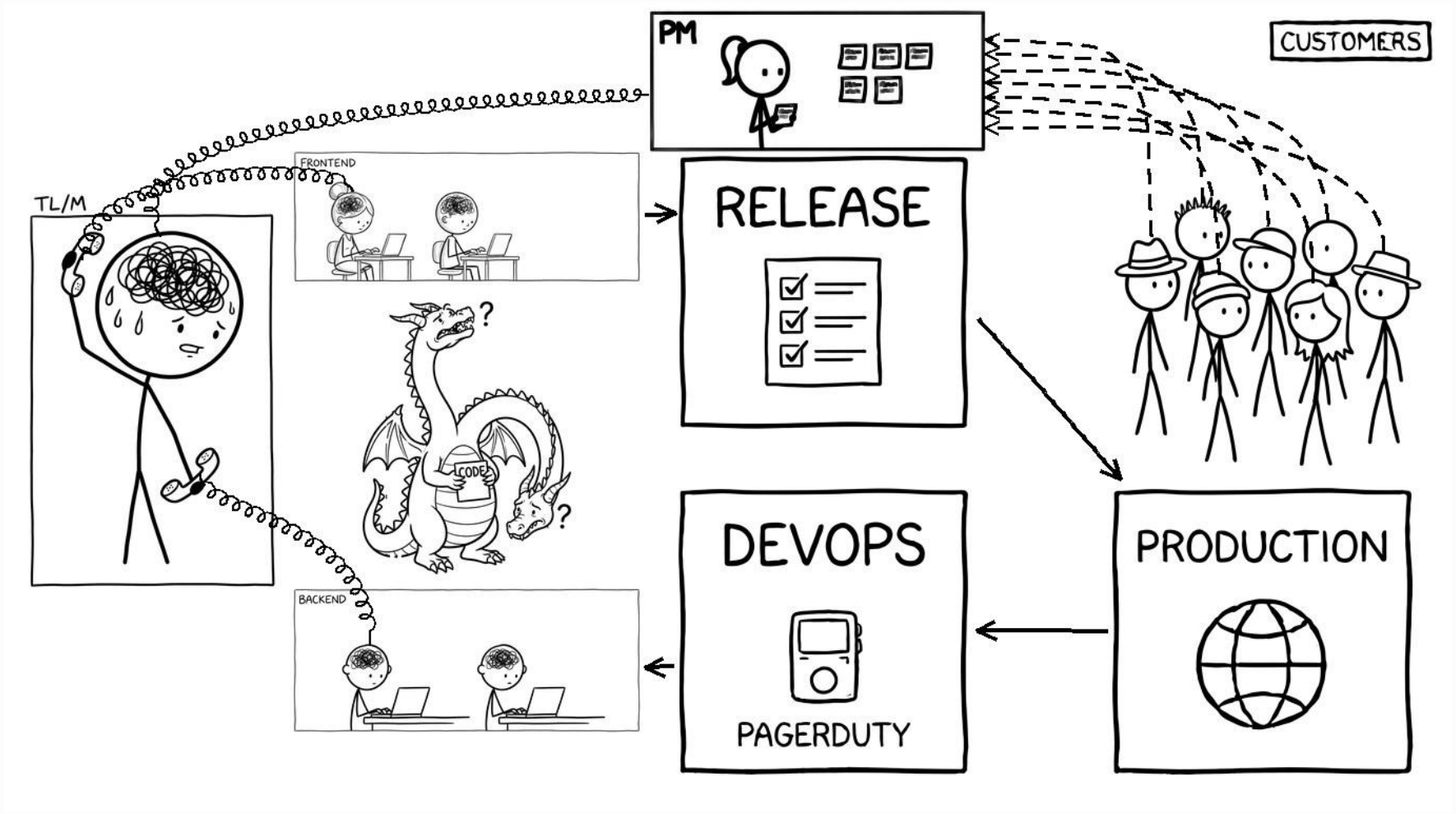
TWO ENGINEERS ≠ TWICE AS FAST.

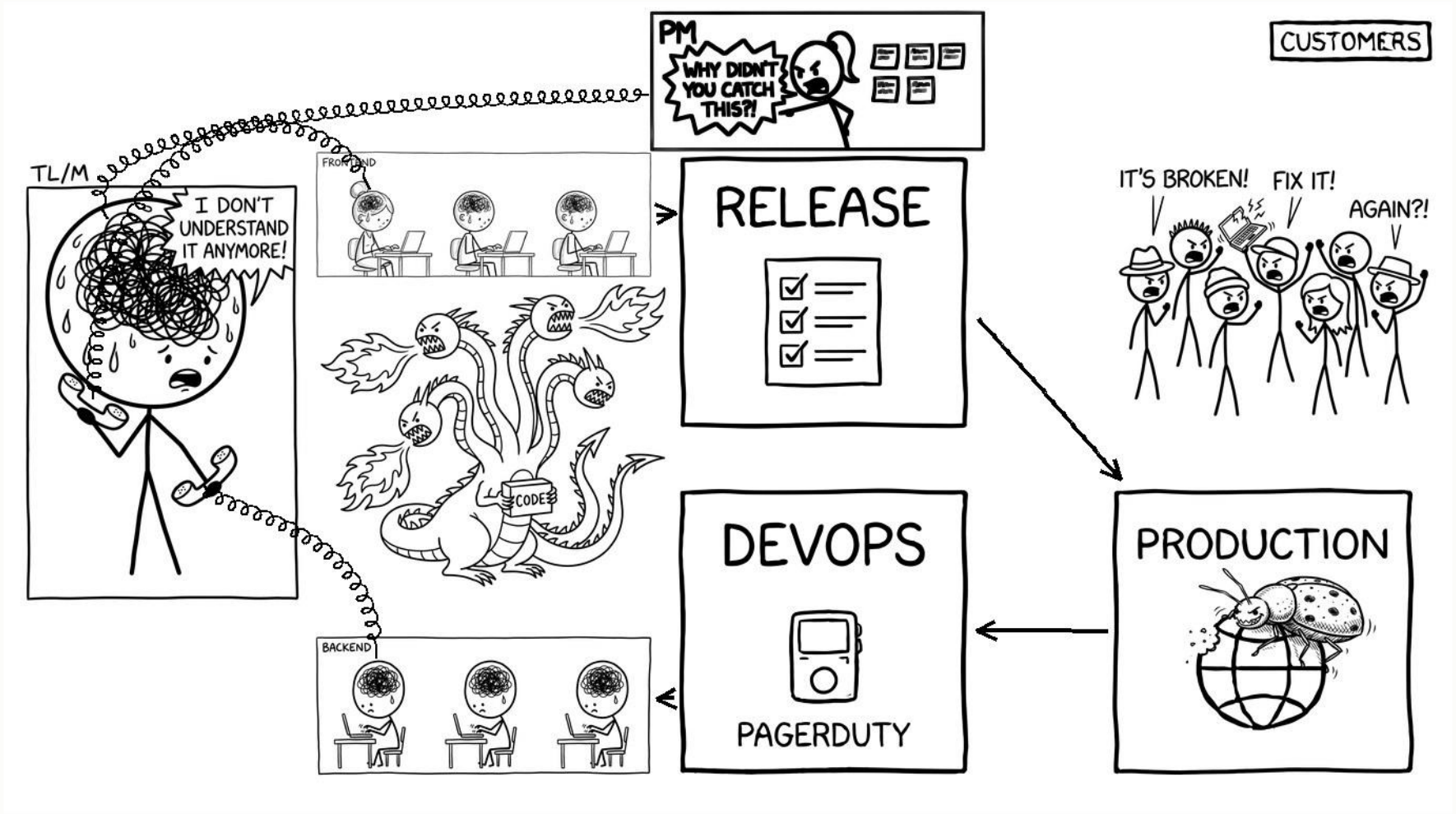

CUSTOMERS
TEAM
PM-LIGHT
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

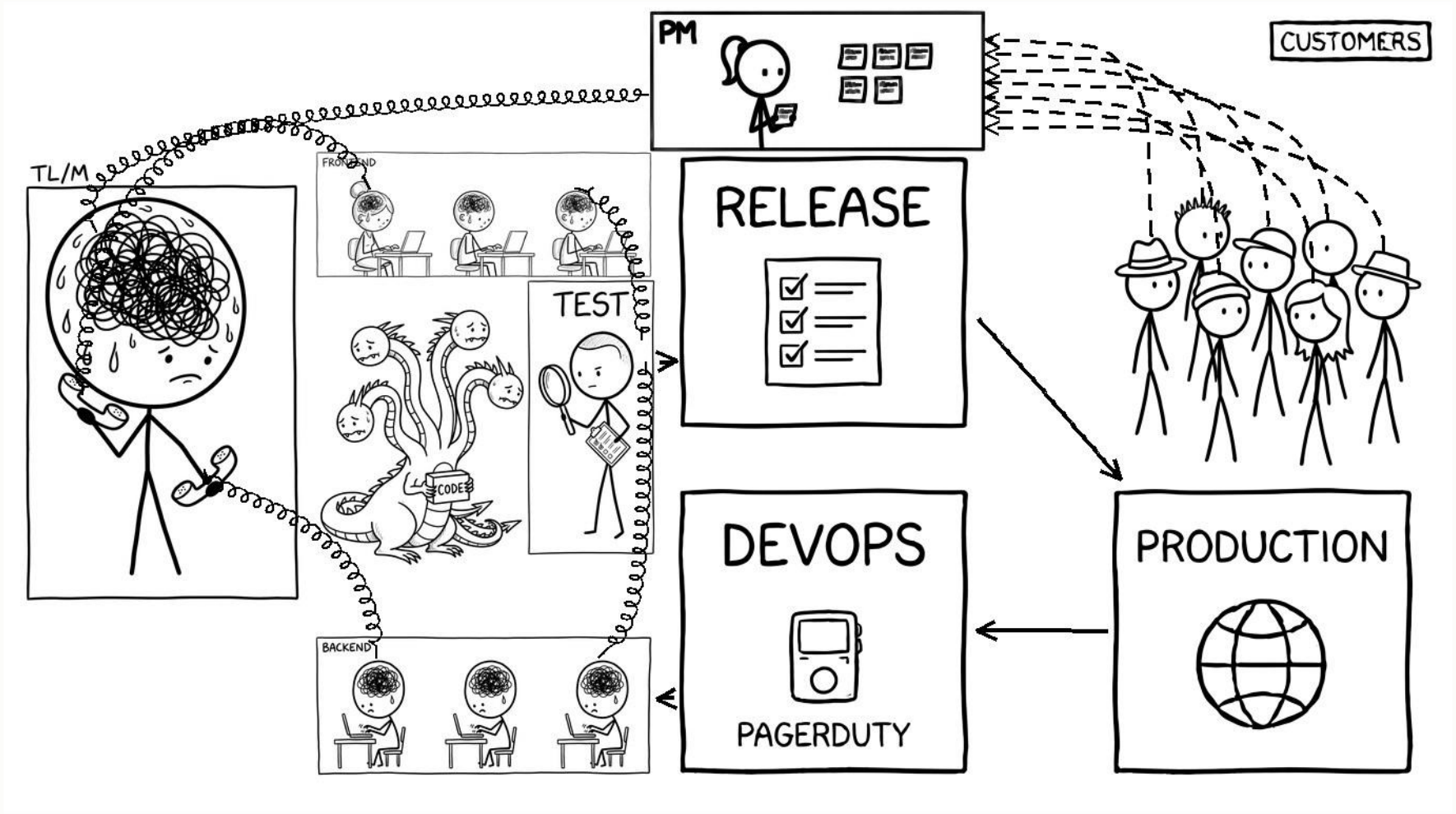

PM
CUSTOMERS
TEAM
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

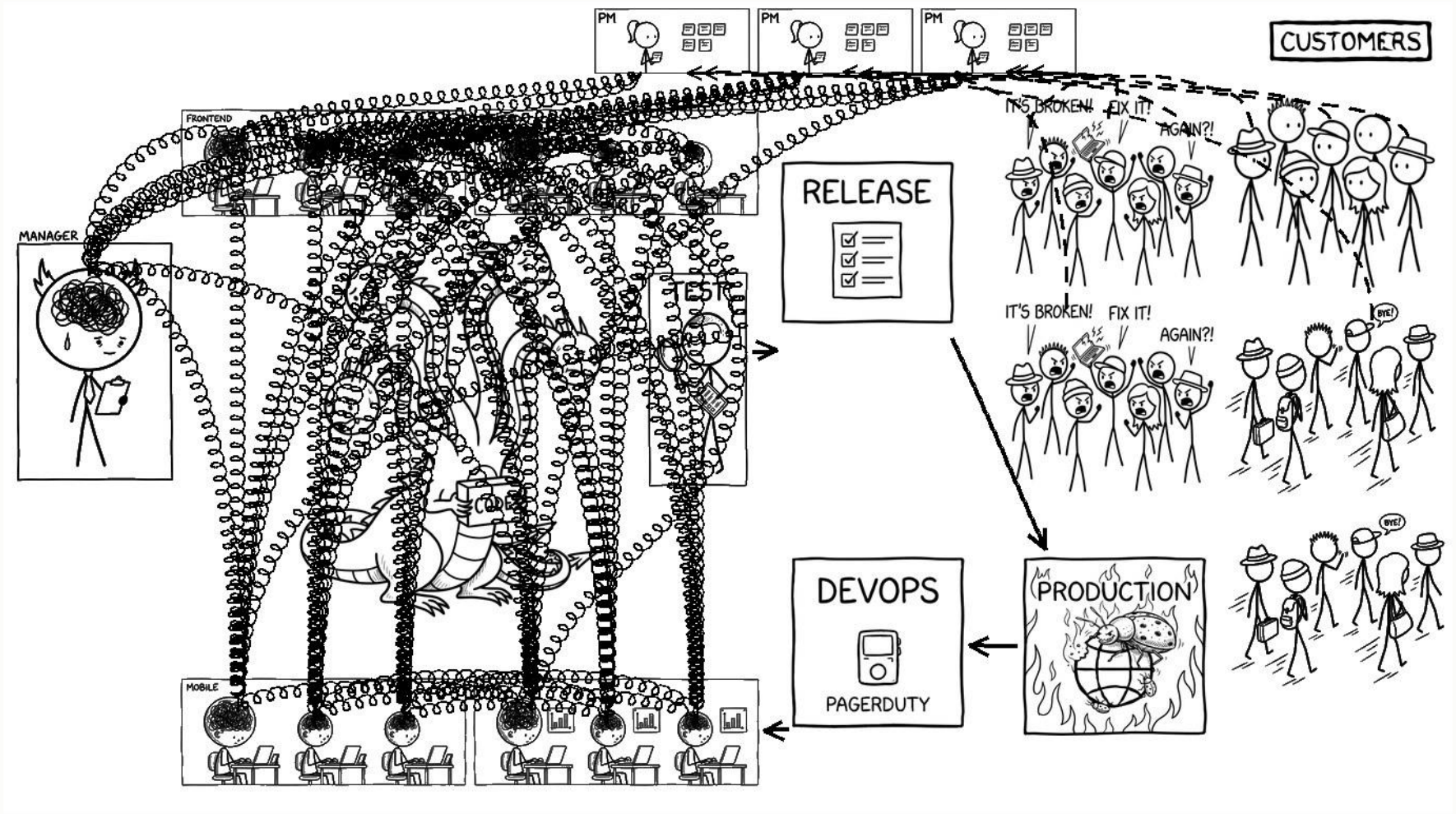

PM
CUSTOMERS
TEAM
RELEASE
CAN IT DO X?
I NEED Y!
WHAT ABOUT Z?
ALSO...
CODE
DEVOPS
PAGERDUTY
PRODUCTION

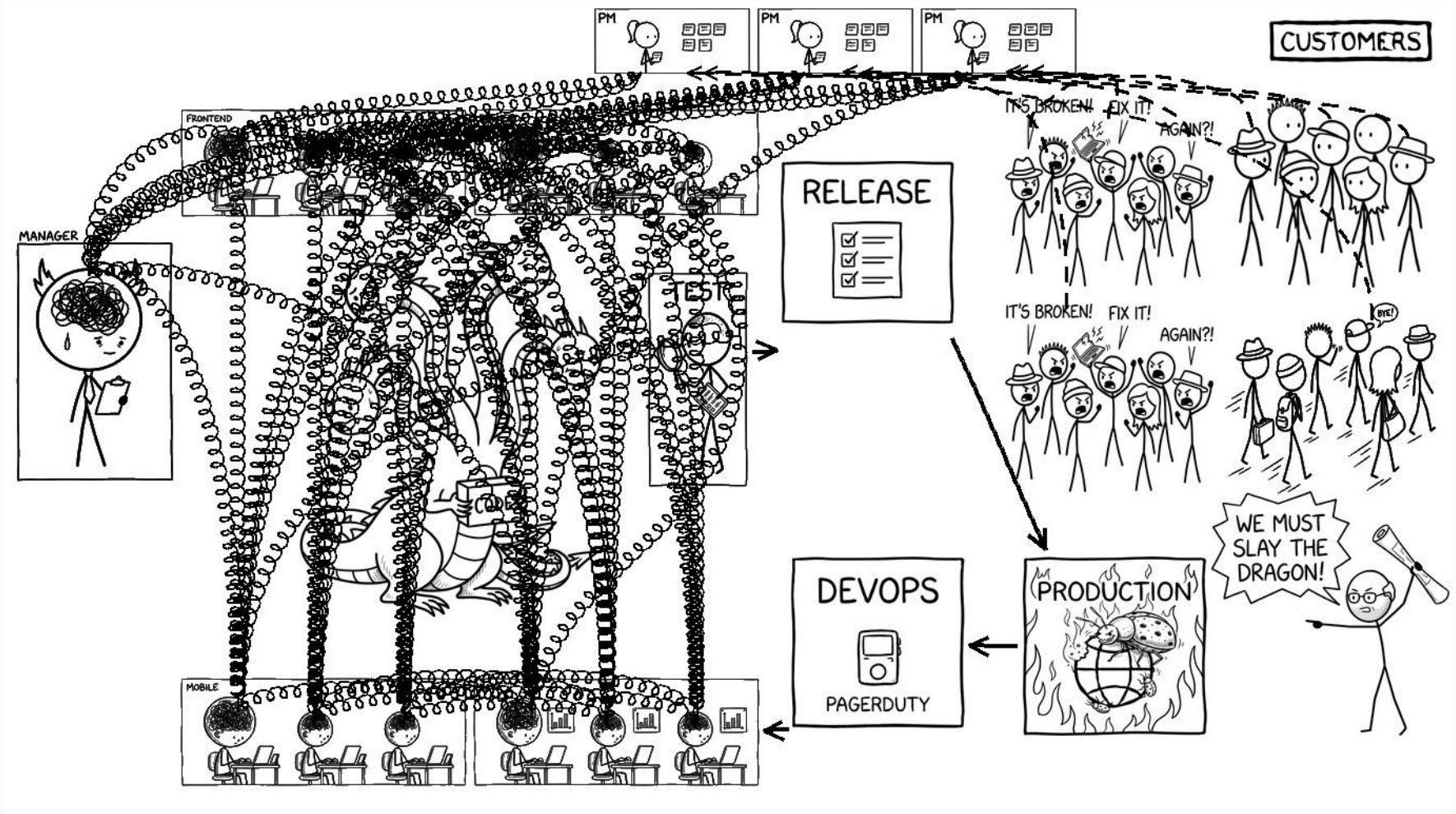

PM
PROJECT PLAN
CUSTOMERS
TEAM
CODE
RELEASE
DEVOPS
PAGERDUTY
PRODUCTION

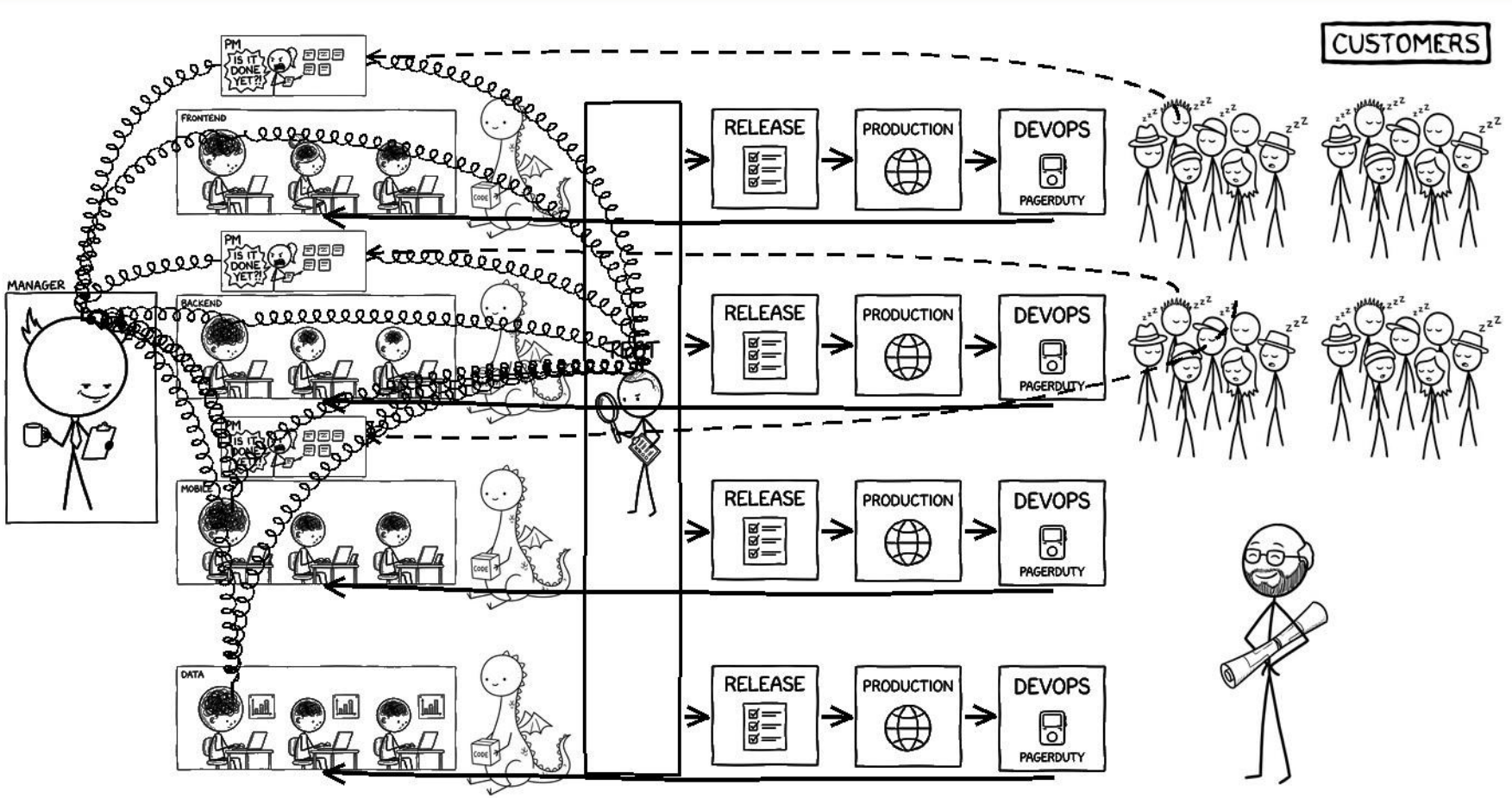

PM
CUSTOMERS
TL/M
FRONTEND
RELEASE
CODE
BACKEND
DEVOPS
PAGERDUTY
PRODUCTION

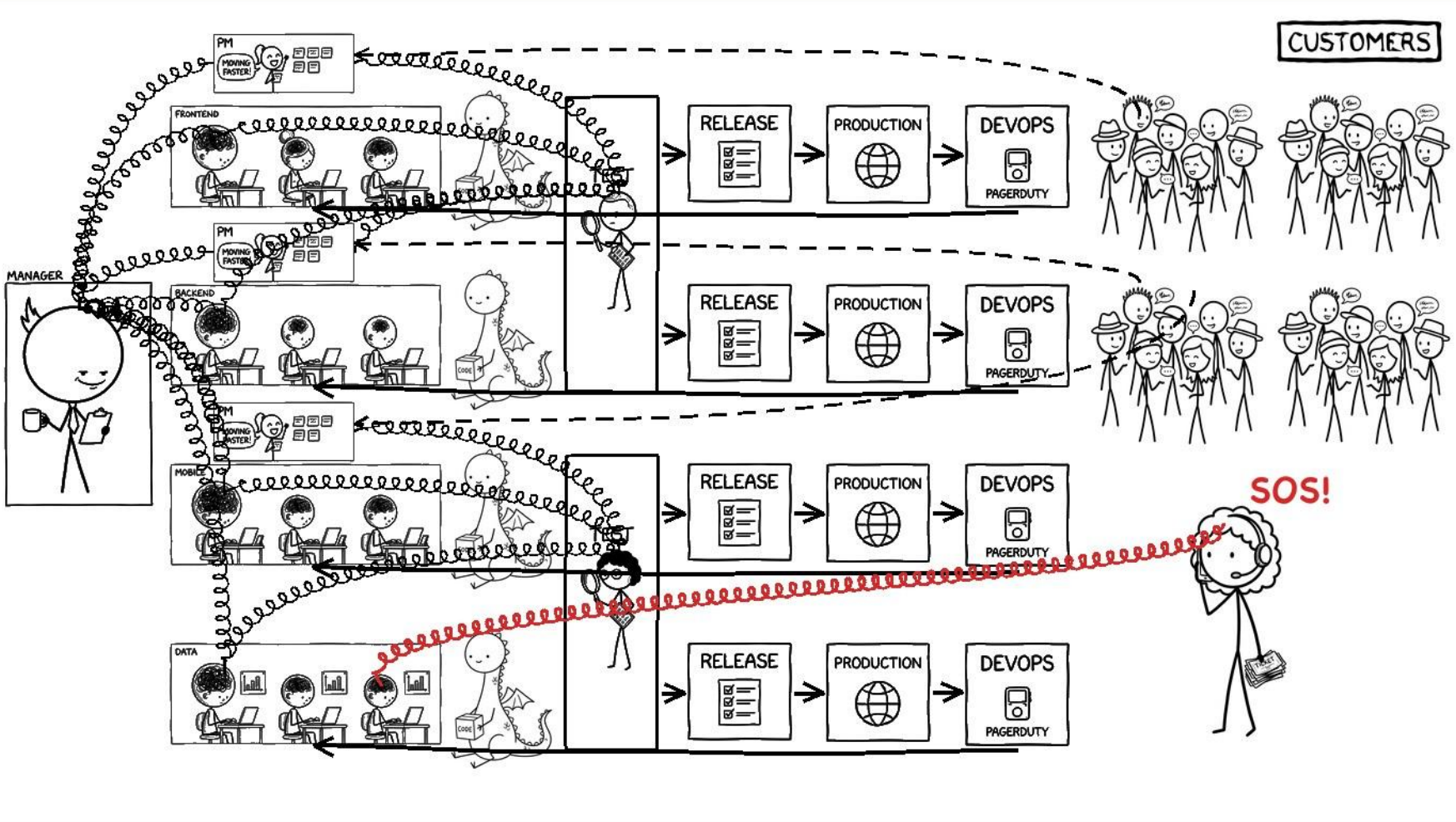

PM
WHY DIDN'T YOU CATCH THIS?!
CUSTOMERS
TL/M
I DON'T UNDERSTAND IT ANYMORE!
FRONTEND
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
CODE
DEVOPS
PAGERDUTY
PRODUCTION
BACKEND

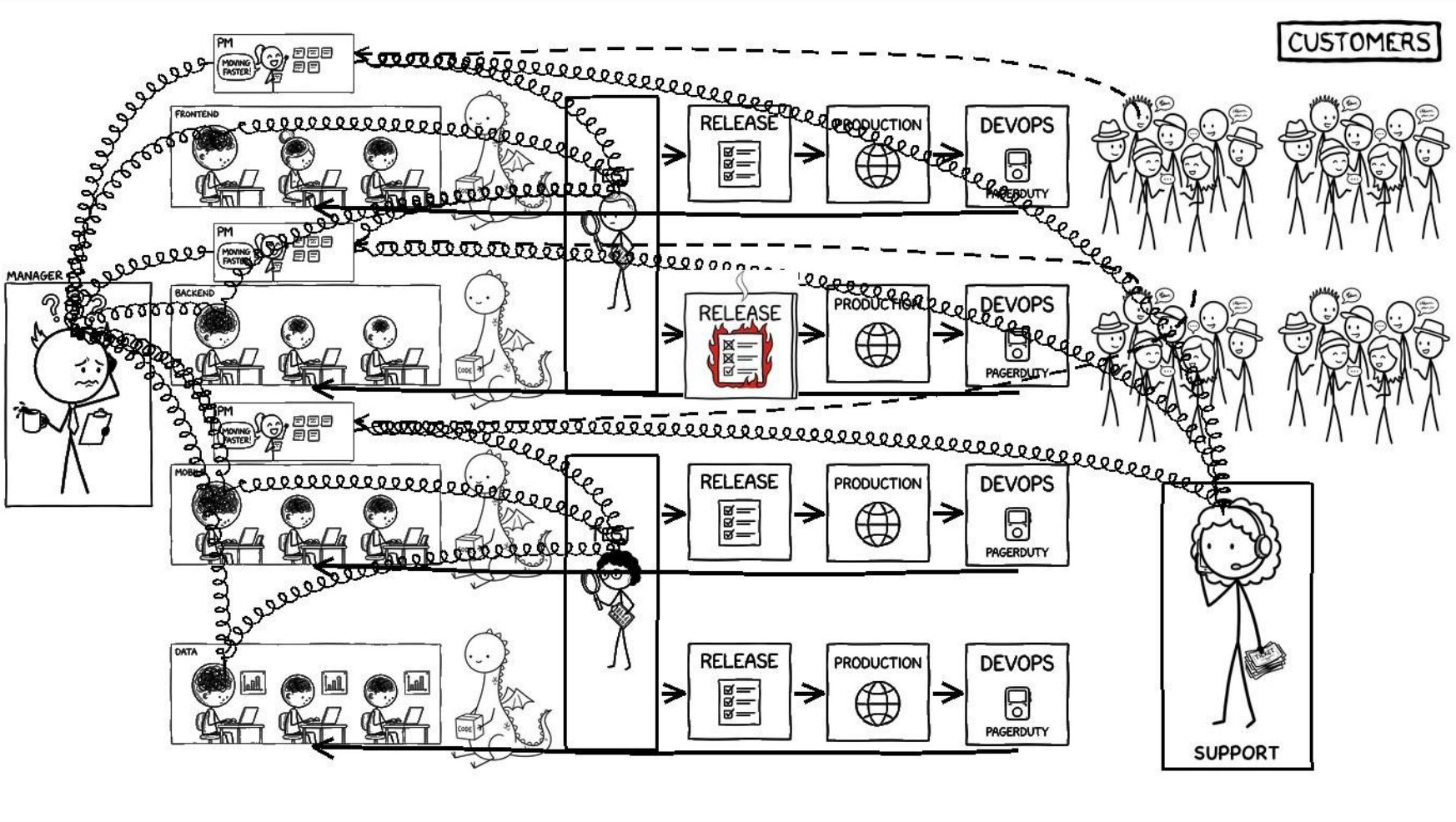

PM
CUSTOMERS
TL/M
FRONTEND
RELEASE
TEST
CODE
DEVOPS
PRODUCTION
BACKEND
PAGERDUTY

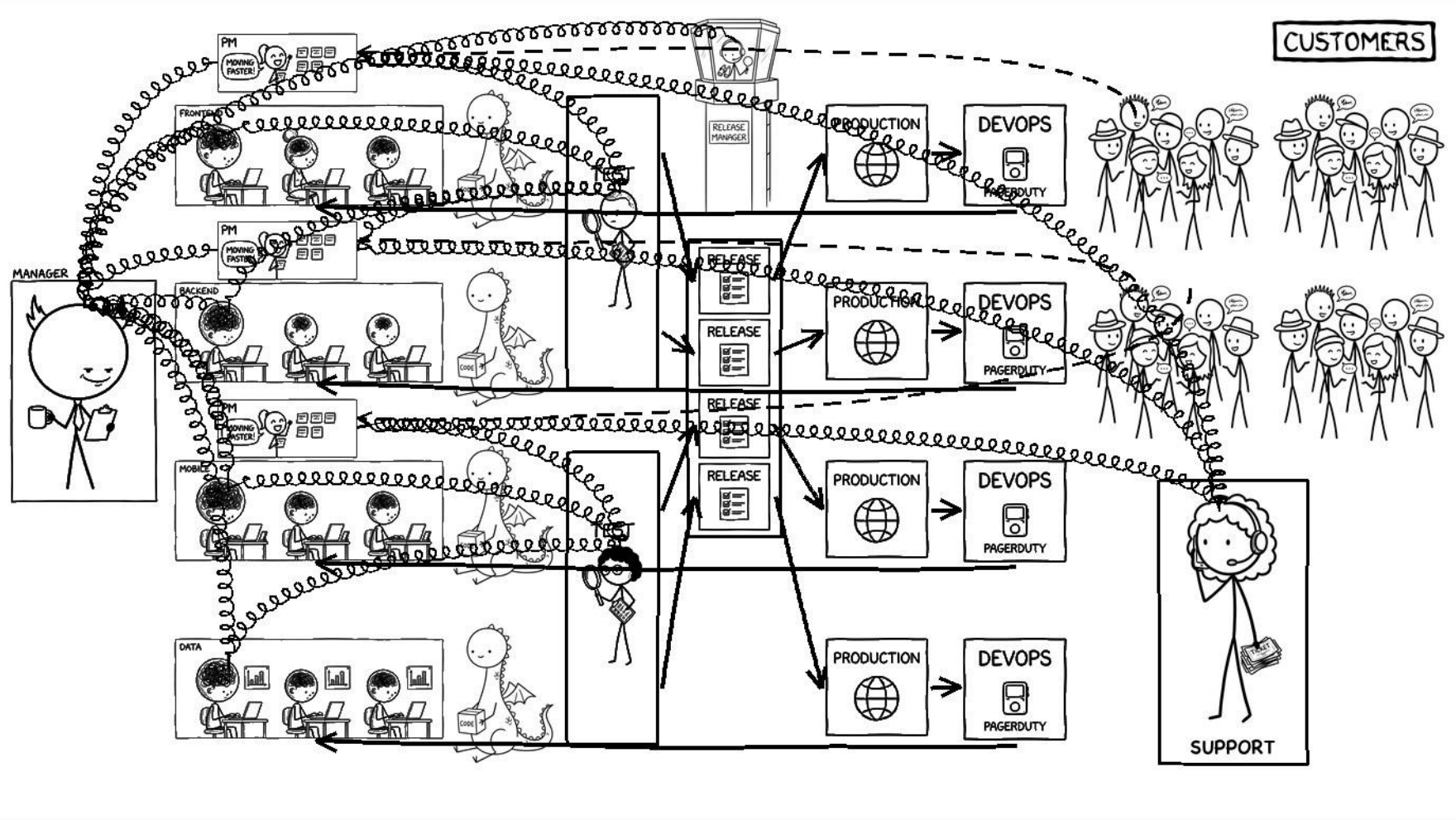

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
BYE!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

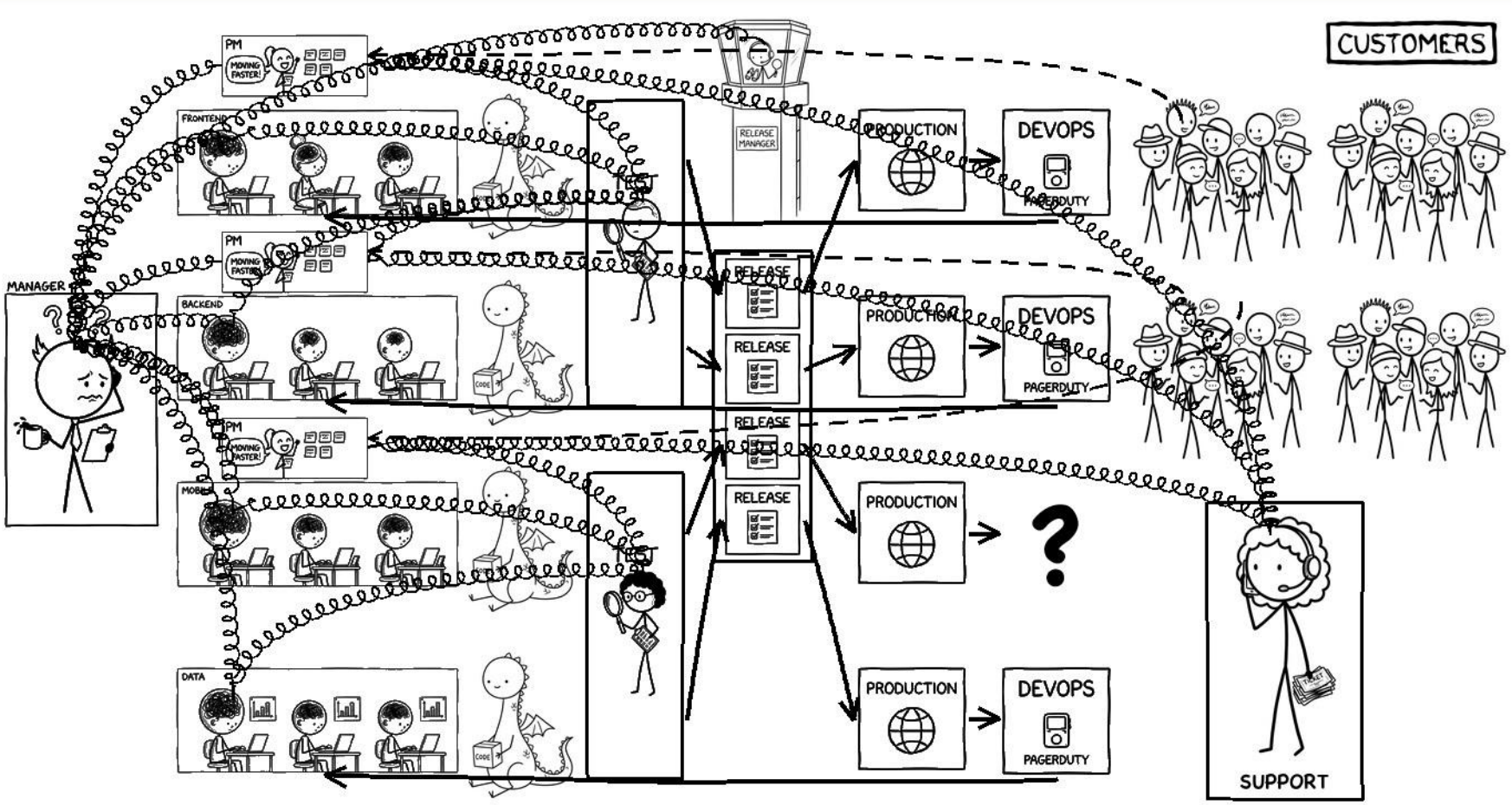

PM
PM
PM
CUSTOMERS
FRONTEND
MANAGER
TEST
RELEASE
IT'S BROKEN!
FIX IT!
AGAIN?!
IT'S BROKEN!
FIX IT!
AGAIN?!
BYE!
WE MUST SLAY THE DRAGON!
DEVOPS
PAGERDUTY
PRODUCTION
MOBILE

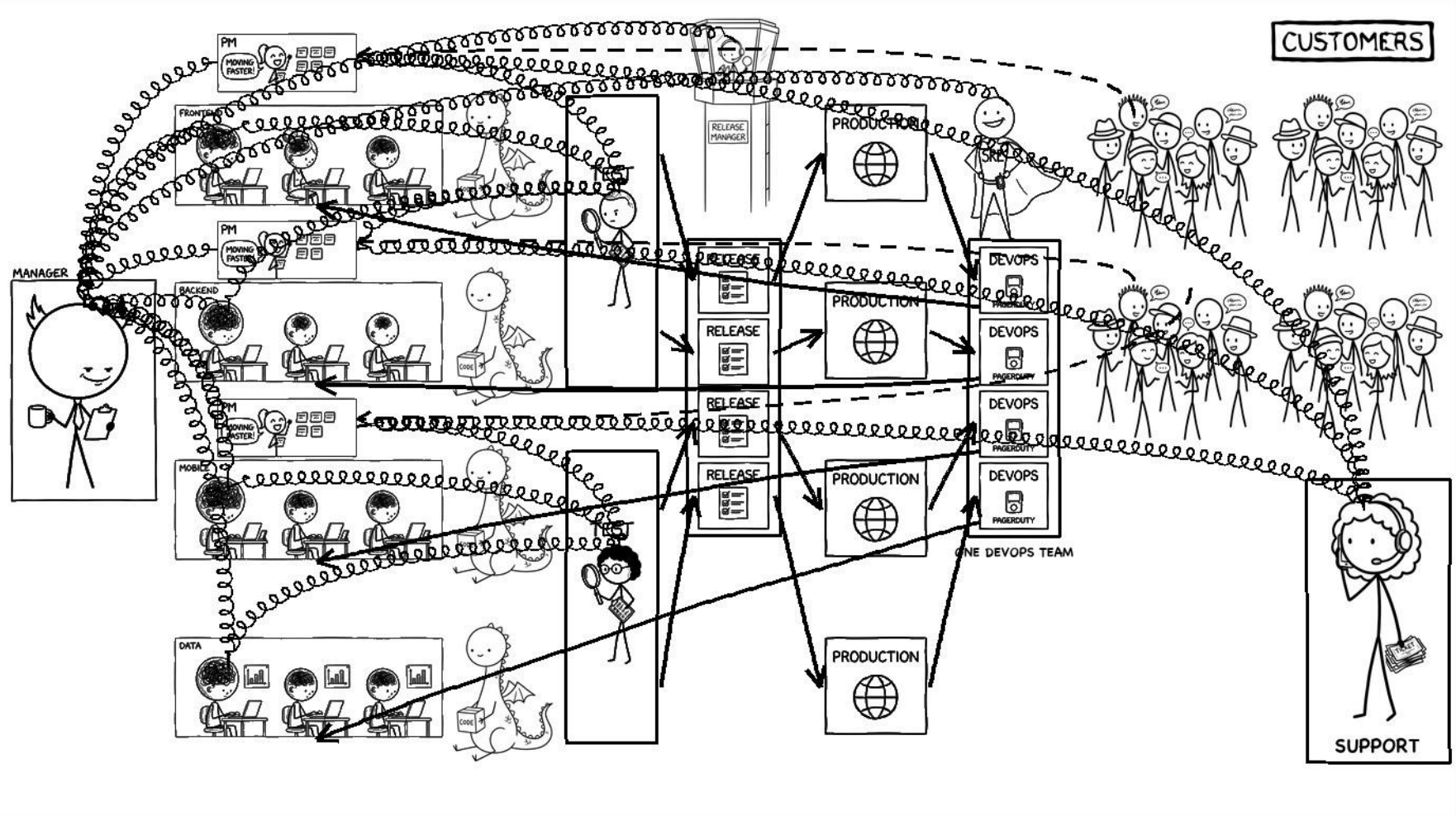

CUSTOMERS
PM
IS IT DONE YET?!
FRONTEND
PM
IS IT DONE YET?!
MANAGER
BACKEND
PM
IS IT DONE YET?!
MOBILE
DATA
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY

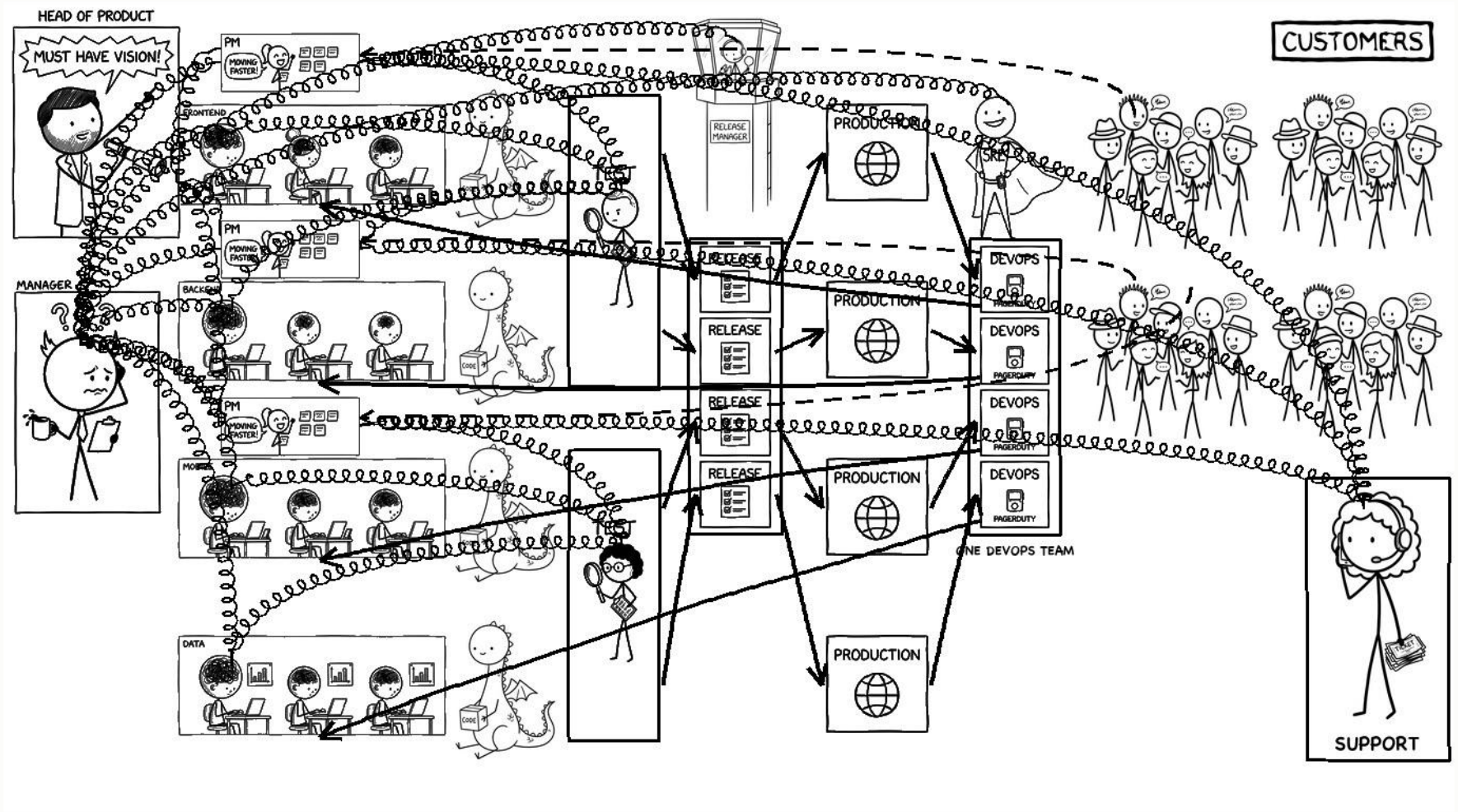

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
CODE
PM
MOVING FASTER!
MOBILE
DATA
CODE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
SOS!


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
BACKEND
MOBILE
DATA
MANAGER
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
SUPPORT
TICKET

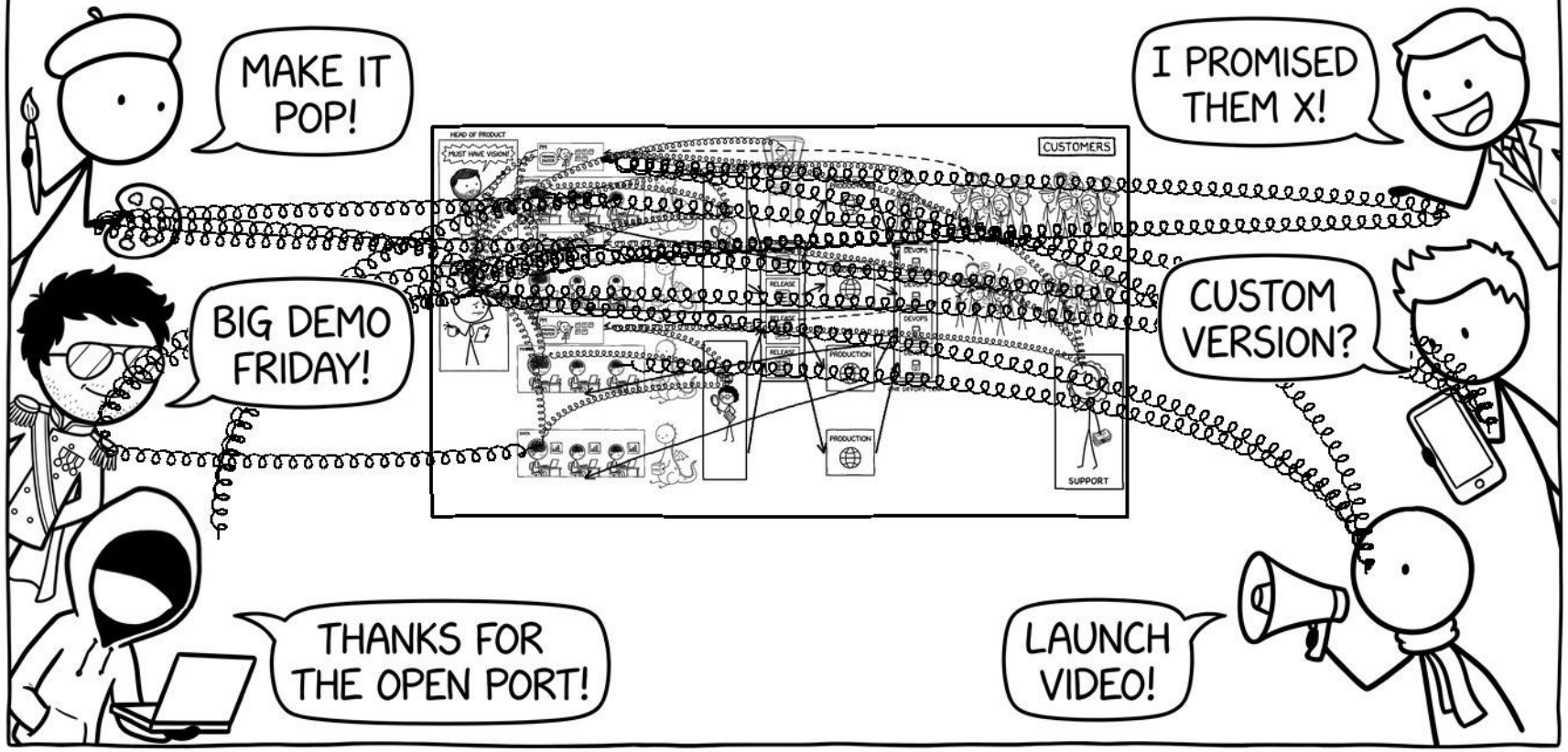

CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
DEVOPS
PAGERDUTY
PRODUCTION
?
PRODUCTION
DEVOPS
PAGERDUTY
SUPPORT


CUSTOMERS
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
SRE
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
SUPPORT


HEAD OF PRODUCT
MUST HAVE VISION!
PM
MOVING FASTER!
FRONTEND
PM
MOVING FASTER!
MANAGER
BACKEND
PM
MOVING FASTER!
MOBILE
DATA
RELEASE MANAGER
RELEASE
RELEASE
RELEASE
RELEASE
PRODUCTION
PRODUCTION
PRODUCTION
PRODUCTION
SRE
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
DEVOPS
PAGERDUTY
ONE DEVOPS TEAM
CUSTOMERS
SUPPORT

OH, AND DON'T FORGET...

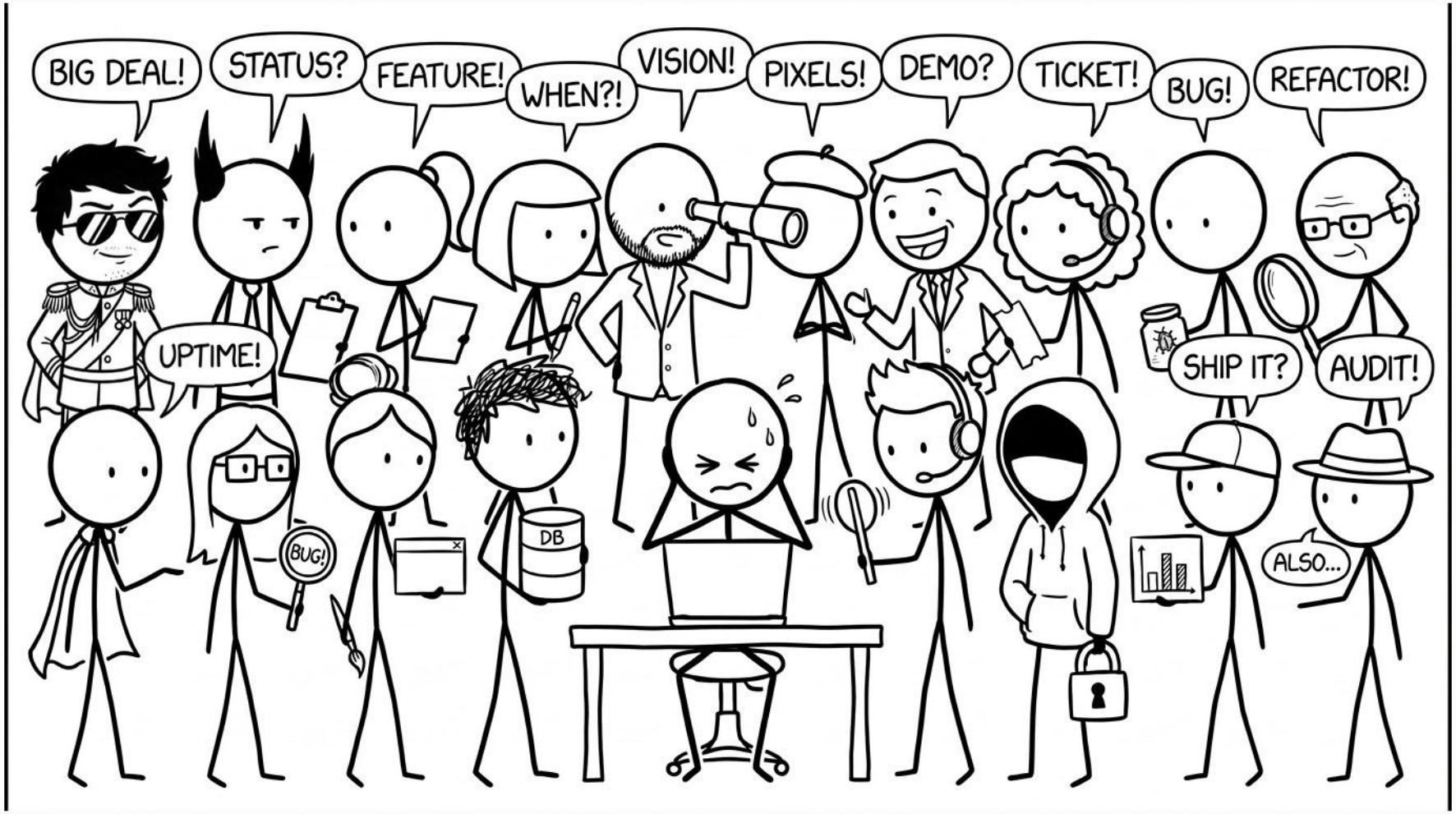


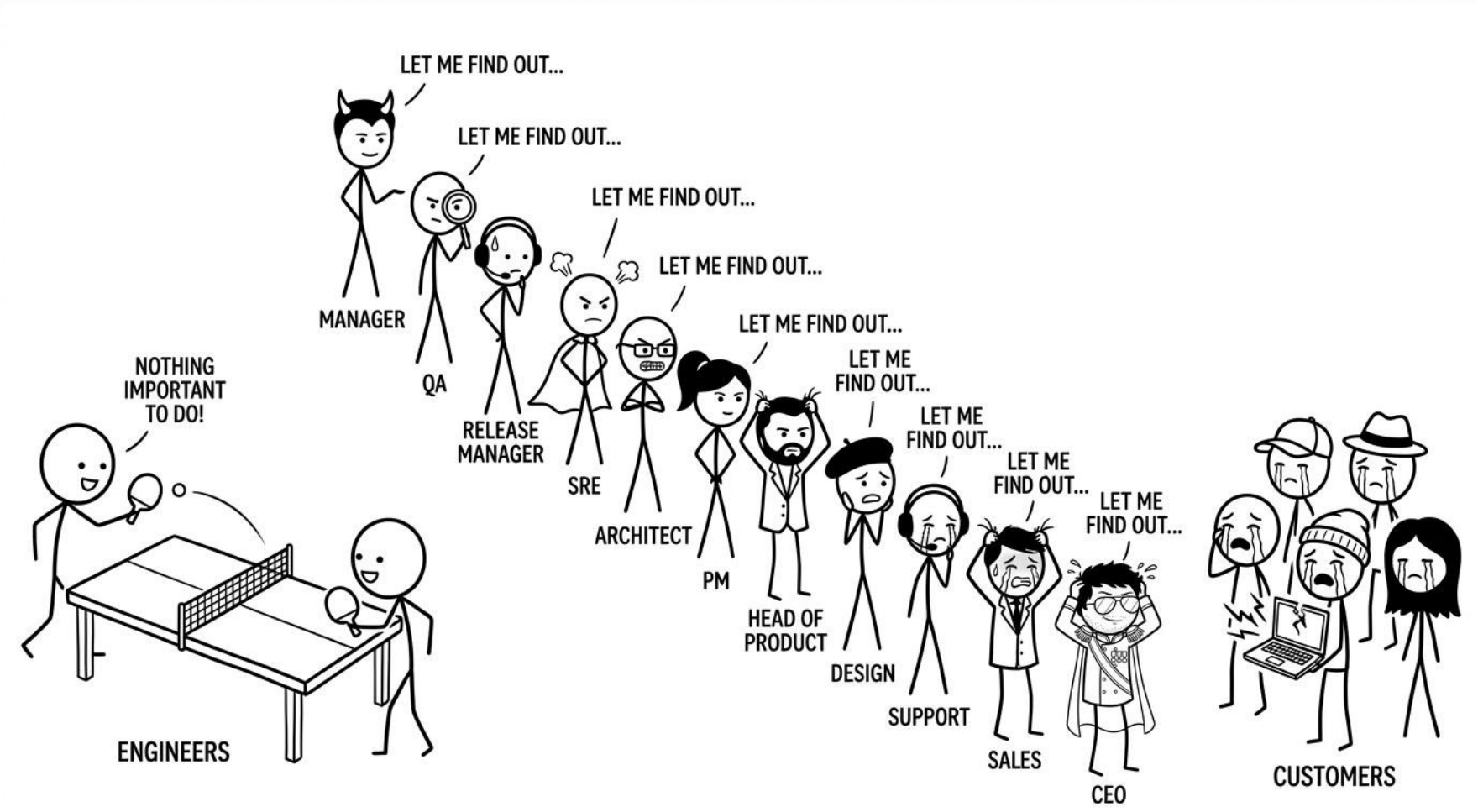
Common Org Failure Modes

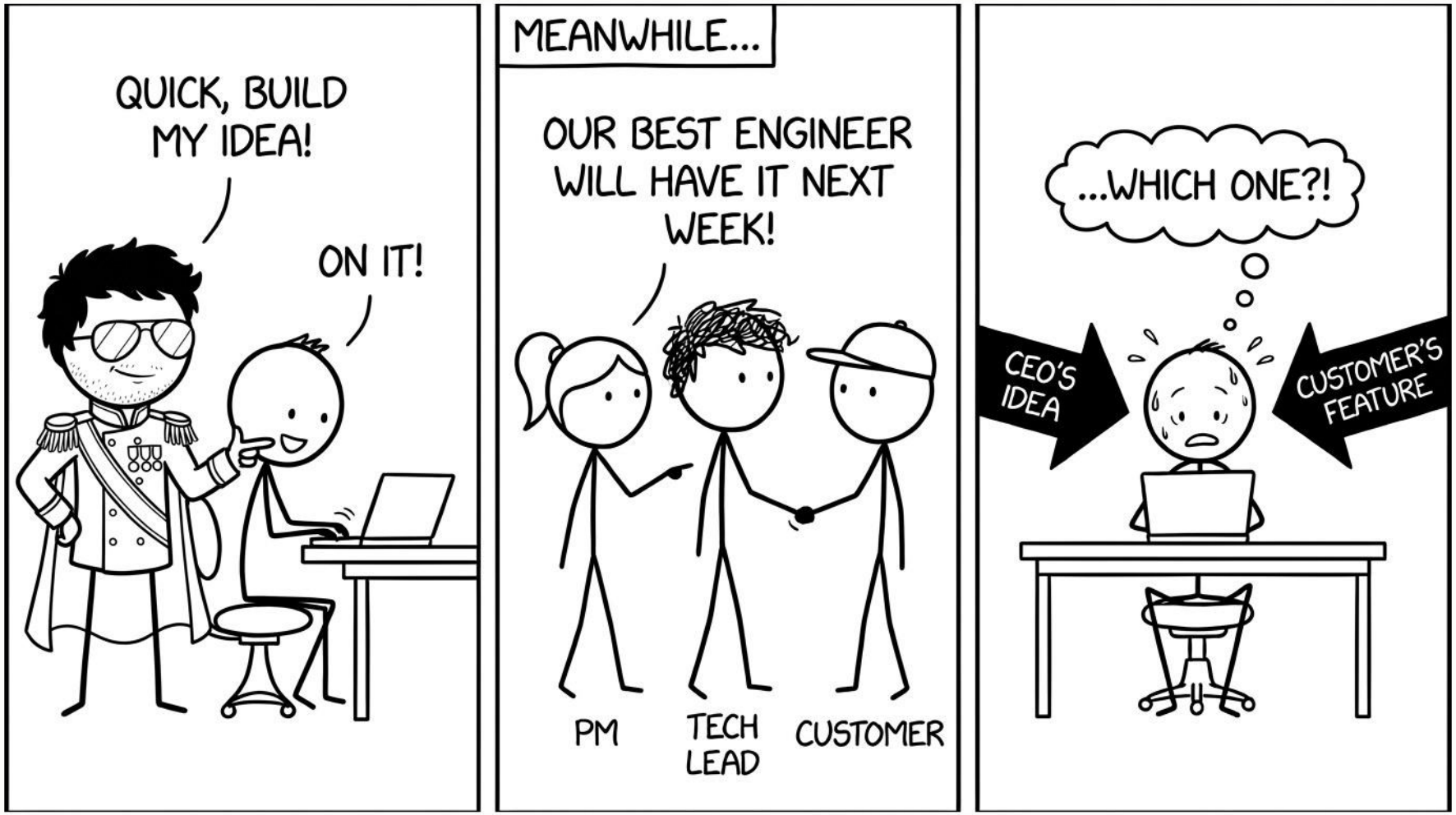
What it looks like when the org around engineering goes wrong

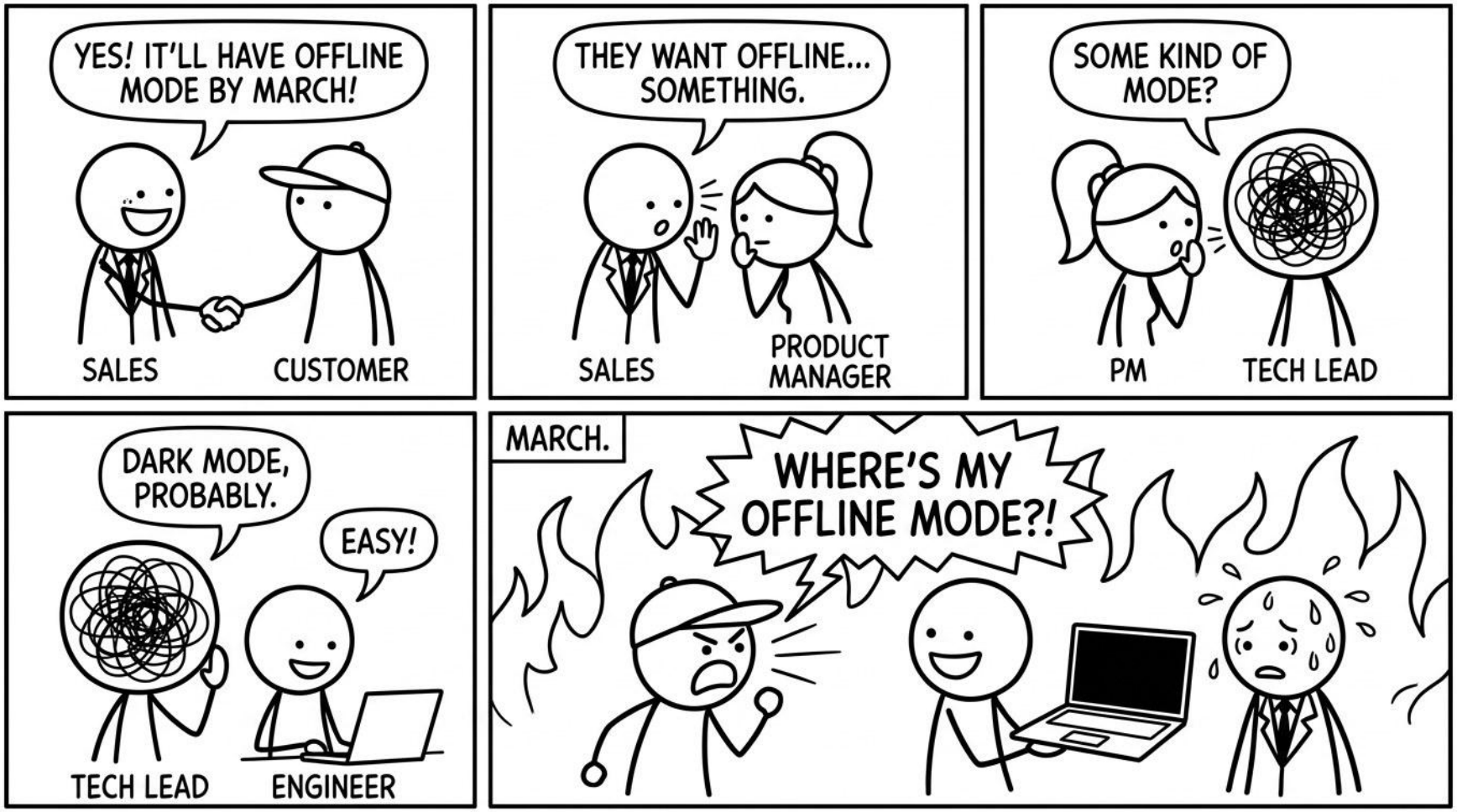


Guess there's nothing to do...


BIG DEAL!
STATUS?
FEATURE!
WHEN?!
VISION!
PIXELS!
DEMO?
TICKET!
BUG!
REFACTOR!
UPTIME!
SHIP IT?
AUDIT!
BUG!
DB
ALSO...


NOTHING IMPORTANT TO DO!
ENGINEERS
LET ME FIND OUT...
MANAGER
LET ME FIND OUT...
QA
LET ME FIND OUT...
RELEASE MANAGER
LET ME FIND OUT...
SRE
LET ME FIND OUT...
ARCHITECT
LET ME FIND OUT...
PM
LET ME FIND OUT...
HEAD OF PRODUCT
LET ME FIND OUT...
DESIGN
LET ME FIND OUT...
SUPPORT
LET ME FIND OUT...
SALES
LET ME FIND OUT...
CEO
CUSTOMERS


QUICK, BUILD MY IDEA!
ON IT!
MEANWHILE...
OUR BEST ENGINEER WILL HAVE IT NEXT WEEK!
PM
TECH LEAD
CUSTOMER
...WHICH ONE?!
CEO'S IDEA
CUSTOMER'S FEATURE


YES! IT'LL HAVE OFFLINE MODE BY MARCH!
SALES
CUSTOMER
THEY WANT OFFLINE... SOMETHING.
SALES
PRODUCT MANAGER
SOME KIND OF MODE?
PM
TECH LEAD
DARK MODE, PROBABLY.
EASY!
TECH LEAD
ENGINEER
MARCH.
WHERE'S MY OFFLINE MODE?!


REFACTOR FIRST!
FEATURE X!
BUG #42!
THE BIG CLIENT!
TOP PRIORITY
#4021
#4021
MUNCH MUNCH
POPCORN